COURSE SLIDES · WSQ

WSQ Technical Drawing with AutoCAD

WSQ Course Code: TGS-2021005538

Skills Framework TSC: Technical Drawing (DSN-TDR-3005-1.11)

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v12 · 18 August 2026

© 2026 Tertiary Infotech Academy Pte Ltd. All rights reserved. · www.tertiarycourses.com.sg

COURSE ADMINISTRATION

Welcome & Housekeeping

© 2026 Tertiary Infotech Academy Pte Ltd

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

© 2026 Tertiary Infotech Academy Pte Ltd

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

© 2026 Tertiary Infotech Academy Pte Ltd

YOUR TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

SPECIALISATION
CAD, technical drawing and engineering design — AutoCAD, Revit and Fusion 360.

DELIVERS
WSQ courses on technical drawing, design and emerging technologies.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

© 2026 Tertiary Infotech Academy Pte Ltd

ICE-BREAKER

Let's Know Each Other

- Your name and organisation / role.
- Your experience with CAD, drafting or design (if any).
- What you want to be able to draw or document after this course.

© 2026 Tertiary Infotech Academy Pte Ltd

HOUSEKEEPING

Ground Rules

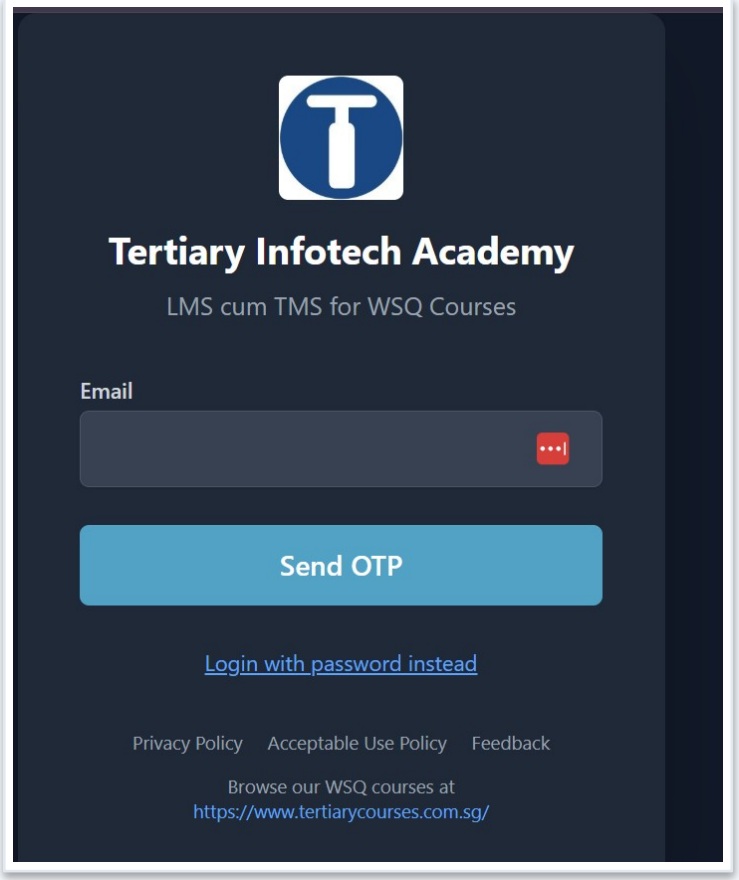
1. Set your mobile phone to silent mode.
2. Participate actively — no question is stupid.
3. Mutual respect: agree to disagree.
4. One conversation at one time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

© 2026 Tertiary Infotech Academy Pte Ltd

COURSE PORTAL · LMS/TMS

Download Course Material

- Log in to the LMS/TMS portal with the account created at registration.
- Portal: https://lms-tms.tertiaryinfotech.com
- Open your course → Course Materials to download the slides and Learner Guide.
- Keep them handy — the final assessment is open book (slides, Learner Guide and approved materials only).



© 2026 Tertiary Infotech Academy Pte Ltd

SCHEDULE

Lesson Plan — 2 Days, 8 hours/day

Day 1

- Day 1 — AutoCAD Foundations, Drawing & Modifying Objects
  - Digital attendance (AM) · introductions · outcomes
  - Topic 1: Getting Started with AutoCAD (Lab 1)
  - Topic 2: Technical Drawing with AutoCAD (Labs 2–5)
  - Topic 3: Modifying Objects (Labs 6–8)

Day 2

- Day 2 — Organizing Objects, Design Documentation & Assessment
  - Topic 4: Organizing Objects (Labs 9–12)
  - Topic 5: Design Documentation (Labs 13–16)
  - Course feedback & TRAQOM survey
  - Final Assessment: WA (SAQ) + Practical Performance
  - Daily timing: 9:30am–6:30pm · 1-hour lunch

© 2026 Tertiary Infotech Academy Pte Ltd

WSQ ALIGNMENT

Skills Framework — TSC

1

TSC Title

Technical Drawing

2

TSC Code

DSN-TDR-3005-1.11

3

Framework

Skills Framework for Design — Technical Drawing, Proficiency Level 3.

4

Coverage

Five abilities (A1–A5) and five knowledge areas (K1–K5) mapped to the five course topics.

© 2026 Tertiary Infotech Academy Pte Ltd

SKILLS FRAMEWORK TSC

TSC Abilities & Knowledge

Abilities

- A1: Read and visualise technical drawings according to briefs
- A2: Produce two-dimensional (2D) and three-dimensional (3D) technical drawings based on input from stakeholders
- A3: Create part lists for new design to facilitate technical drawing
- A4: Oversee the development and construction of new designs' prototypes, in collaboration with organisation's production function
- A5: Produce drawings and specifications, that conform to the requirements of relevant standards for submission

Knowledge

- K1: Symbols, standards and conventions used in technical drawings within one's field and region
- K2: Methods of technical drawing by instruments and computer-aided design (CAD) systems
- K3: Types of technical drawings
- K4: Fundamentals of drafting design documentation
- K5: Concepts of information technology (IT) to facilitate the usage of technical drawing software

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 — Read

Read and visualize AutoCAD technical drawings.

2

LO2 — Produce

Produce 2D and 3D technical drawings using AutoCAD.

3

LO3 — Part lists

Create part lists using AutoCAD blocks, attributes and tables.

4

LO4 — Document

Create AutoCAD design documentation — text, dimensions, annotation.

5

LO5 — Specify

Produce design specification: layouts, viewports and plotted sheets.

© 2026 Tertiary Infotech Academy Pte Ltd

FIVE TOPICS · 16 HANDS-ON LABS

Course Outline

Day 1

- Topic 1: Getting Started with AutoCAD (A1, K1)
  - AutoCAD user interface
  - Workspace settings, navigation and units
- Topic 2: Technical Drawing with AutoCAD (A2, K2)
  - Basic AutoCAD operations
  - Drawing objects and isometric drawing
- Topic 3: Modifying Objects (A3, K3)
  - Working with modifiers: move, copy, trim, arrays…

Day 2

- Topic 4: Organizing Objects (A4, K4)
  - Object properties and layers
  - Blocks, attributes and reusable content
- Topic 5: Design Documentation (A5, K5)
  - Text, dimensions and part-list tables
  - Layouts, printing and outputs
  - Annotative techniques

© 2026 Tertiary Infotech Academy Pte Ltd

ASSESSMENT RULES

Briefing for Assessment

- Place phones and other materials under the table or on the floor.
- No photos or recording of assessment scripts.
- No discussion during the assessment.
- Use a black/blue pen for hard-copy assessments.
- No liquid paper / correction tape.
- Scripts are collected when time is up.

© 2026 Tertiary Infotech Academy Pte Ltd

Final Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — produce a dimensioned AutoCAD technical drawing, 1.5 hours, open book.
- Format: Open Book — slides, Learner Guide and approved materials only.
- A minimum of 75% attendance (SSG Digital Attendance record) and a 'Competent' assessment result are required for funding.
- An appeal process is available if required.

© 2026 Tertiary Infotech Academy Pte Ltd

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then Practical Performance — open book
4. Submit your answers and DWG file on the LMS
5. Sign the Assessment Summary Record

© 2026 Tertiary Infotech Academy Pte Ltd

CORE CONCEPTS

Technical Drawing & CAD Fundamentals

© 2026 Tertiary Infotech Academy Pte Ltd

What is a Technical Drawing?

1

Precise communication

A technical drawing communicates a design's exact geometry, sizes and finishes for manufacture or construction.

2

Standards-based

Symbols, line conventions, projections and dimensioning follow standards (ISO/ANSI) so any reader interprets it the same way.

3

Types

Orthographic multi-view, isometric/pictorial, section, detail and assembly drawings — each answers a different question.

4

CAD advantage

AutoCAD makes drawings precise, editable, layered, reusable and printable at any scale.

© 2026 Tertiary Infotech Academy Pte Ltd

METHODOLOGY

The Technical Drawing Workflow

Set up & draw

- Template, units and limits
- Precision aids: snap, ortho, polar
- Draw geometry at 1:1

Organise & refine

- Layers, properties, blocks
- Modify: trim, fillet, array
- Attributes for part data

Document & publish

- Dimensions, text and tables
- Layouts with scaled viewports
- Plot to PDF for submission

© 2026 Tertiary Infotech Academy Pte Ltd

01

TOPIC 01

Getting Started with AutoCAD

AutoCAD User Interface · Workspace Settings · Navigation · Units

© 2026 Tertiary Infotech Academy Pte Ltd

SKILLS FRAMEWORK A1 · K1

Key Concepts — Getting Started with AutoCAD

1

User interface

Application menu, Quick Access Toolbar, ribbon tabs and panels, drawing area, command window and status bar.

2

Workspaces

Drafting & Annotation, 3D Basics and 3D Modeling — task-oriented sets of ribbons, menus and palettes.

3

Model vs paper space

Design at 1:1 in model space; compose scaled drawing sheets in paper-space layouts.

4

Precision aids

Grid and snap, ortho mode, polar tracking, object snaps and dynamic input keep every pick exact.

5

Navigation

Zoom, pan, ViewCube and the wheel mouse move you around a drawing without changing it.

6

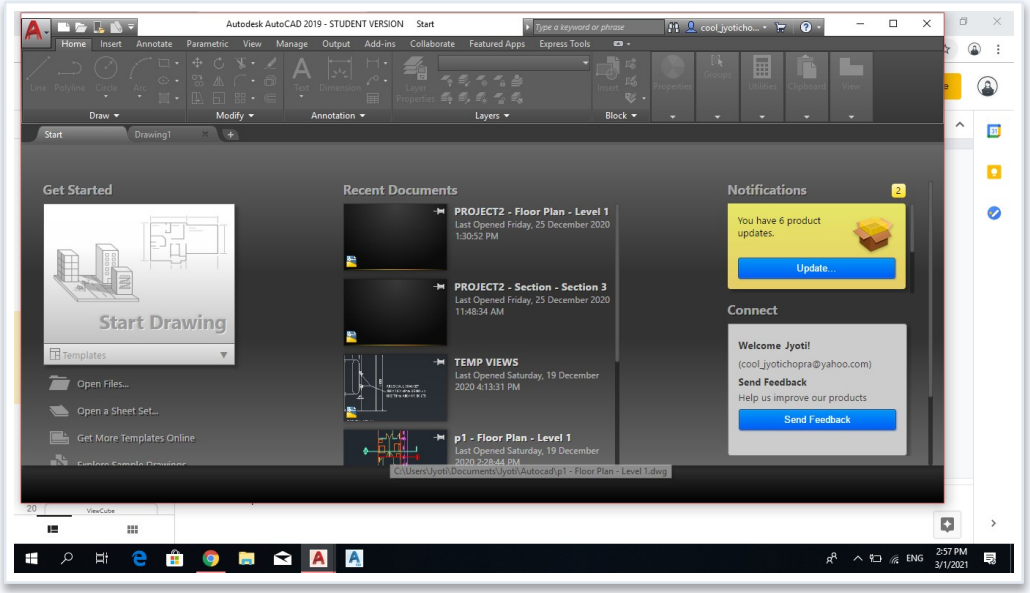
Units and limits

UNITS sets the length/angle format and precision; LIMITS frames the working area of the drawing.

© 2026 Tertiary Infotech Academy Pte Ltd

AutoCAD User Interface — Application Menu

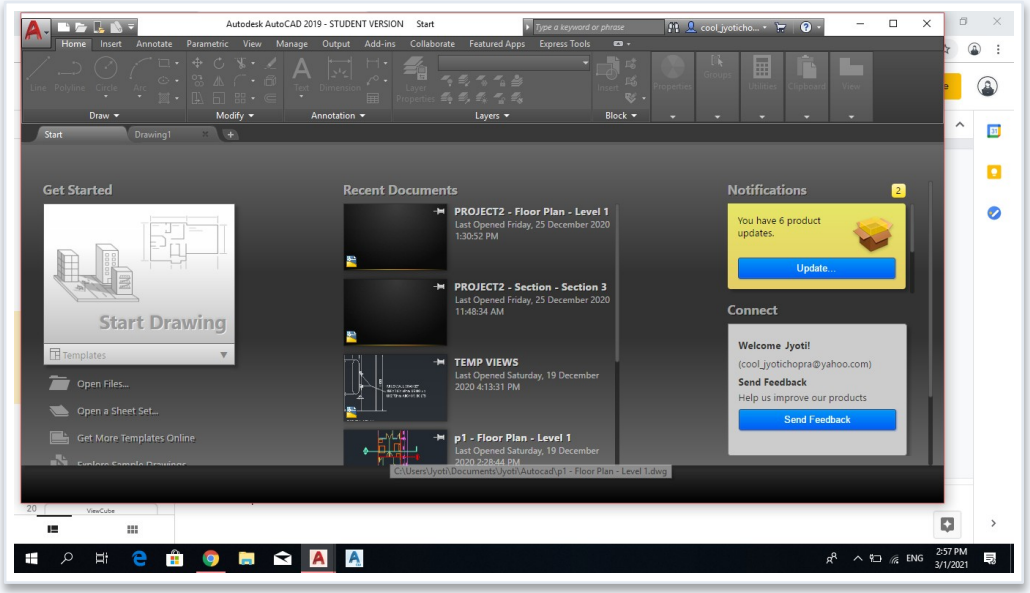
- The Application button opens the file-level menu: create, open and save drawings.
- Audit, recover and purge a drawing file when it needs repair or cleanup.
- Print or publish a file, and access the Options dialog for system settings.
- Recent documents are listed for quick reopening.



© 2026 Tertiary Infotech Academy Pte Ltd

Starting a New Drawing

- Start New Drawing creates a file from the default drawing template (DWT).
- Metric drawings use acadiso.dwt; imperial drawings use acad.dwt.
- The default template is set in the Options dialog.
- Templates carry units, styles, layers and title blocks — start right, draw less.

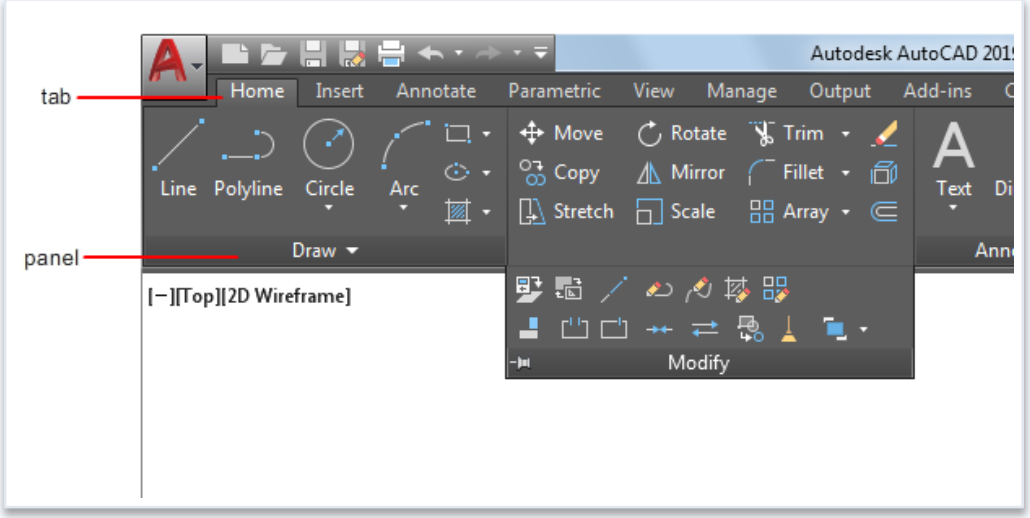


© 2026 Tertiary Infotech Academy Pte Ltd

USER INTERFACE

Ribbon Tabs and Panels

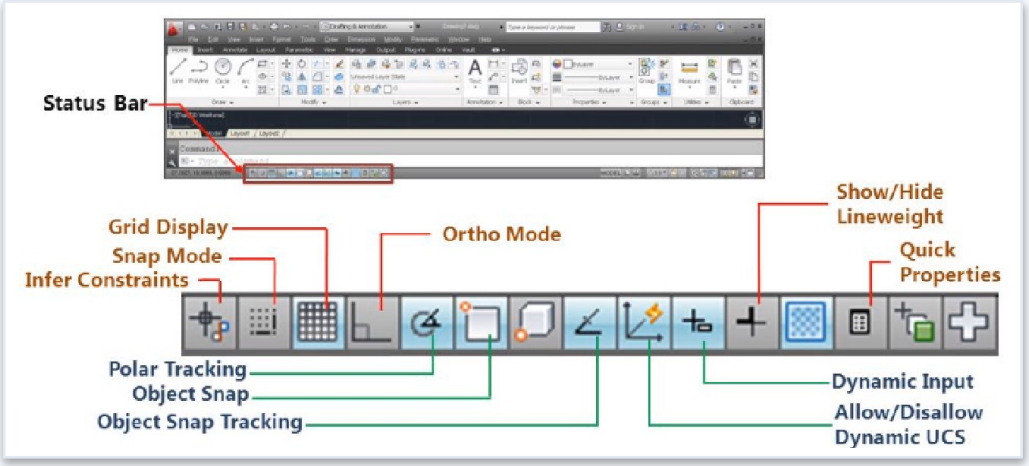
- The ribbon organises tools into tabs (Home, Insert, Annotate, View, Output…).
- Each tab is a set of panels — Draw, Modify, Layers, Annotation and more.
- Panels expand to reveal less-used tools; pin them open while learning.
- Most course commands live on the Home tab's Draw and Modify panels.



© 2026 Tertiary Infotech Academy Pte Ltd

Status Bar — Precision Drafting Aids

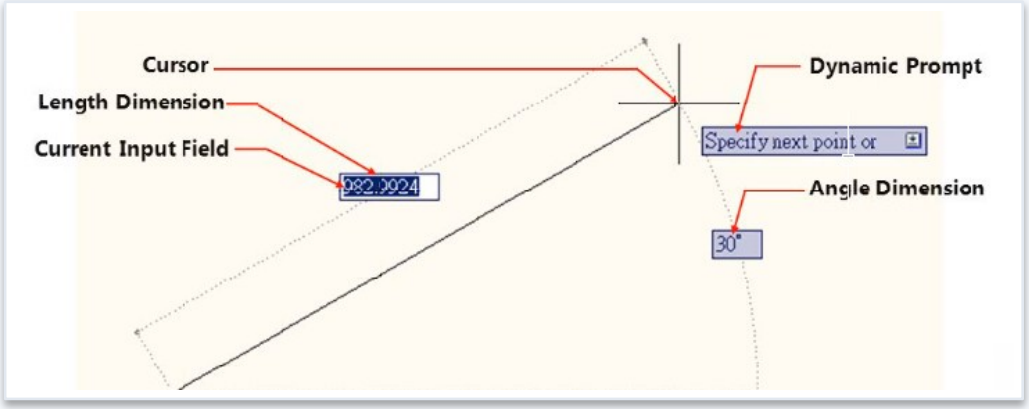
- The status bar toggles grid, snap, ortho, polar tracking and object snap.
- Ortho constrains the cursor to horizontal/vertical; Polar tracks set angles.
- 2D object snap picks exact endpoints, midpoints, centres and intersections.
- Function keys mirror the toggles: F7 grid · F8 ortho · F9 snap · F10 polar · F3 osnap.



© 2026 Tertiary Infotech Academy Pte Ltd

Dynamic Input and the Command Window

- Dynamic input places a command interface beside the cursor as you draw.
- Tooltips show prompts, dimensions and coordinate entry at the crosshair.
- The command window (Ctrl+9) accepts commands and shows every prompt.
- Watch the prompts — AutoCAD always tells you what it expects next.

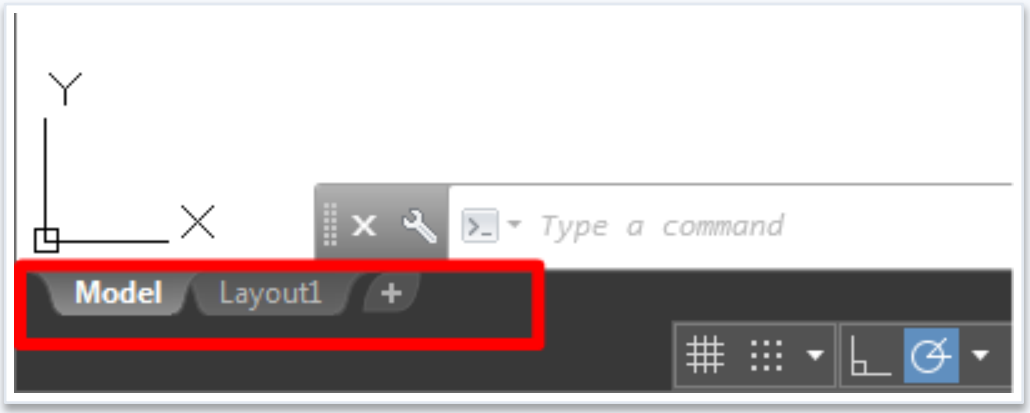


© 2026 Tertiary Infotech Academy Pte Ltd

WORKSPACE

Model Space and Paper Space

- Model space is where you design — always draw at 1:1 real-world size.
- Paper space (layouts) is where you compose scaled sheets for printing.
- The UCS icon shows the origin and orientation of the coordinate system.
- Switch between them with the Model and Layout tabs below the drawing area.

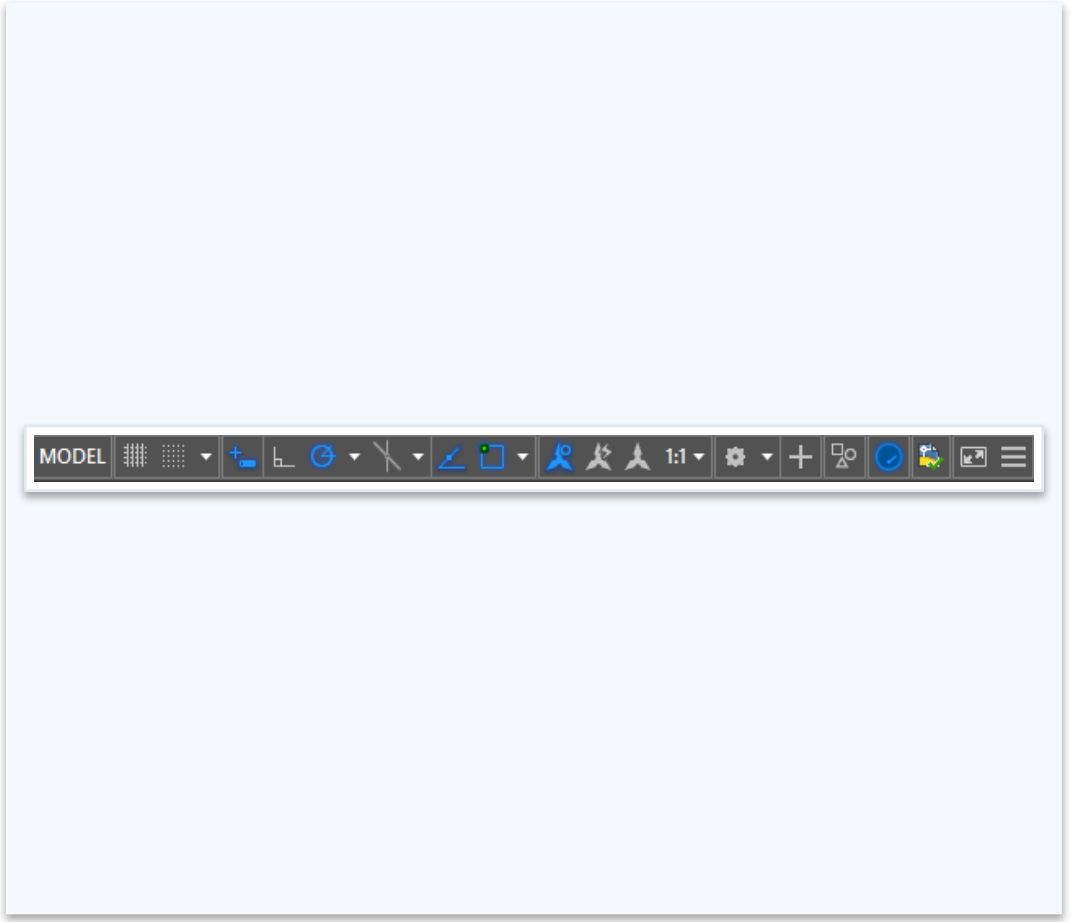


© 2026 Tertiary Infotech Academy Pte Ltd

WORKSPACE

Workspaces and the Quick Access Toolbar

- Workspaces group menus, toolbars and ribbons for a task: Drafting & Annotation, 3D Basics, 3D Modeling.
- This course uses Drafting & Annotation throughout.
- The Quick Access Toolbar sits top-left: New, Open, Save, Undo, Plot.
- Right-click any ribbon tool → Add to Quick Access Toolbar to customise it.



© 2026 Tertiary Infotech Academy Pte Ltd

NAVIGATION

Navigation — Zoom, Pan and the Wheel Mouse

- The navigation bar provides Pan and the full set of Zoom tools.
- ZOOM (Z): Window, All, Extents, Previous, Scale, Object — know them all.
- Wheel mouse: scroll to zoom, press to pan, double-click to zoom extents.
- Navigation changes your view only — it never moves the geometry.

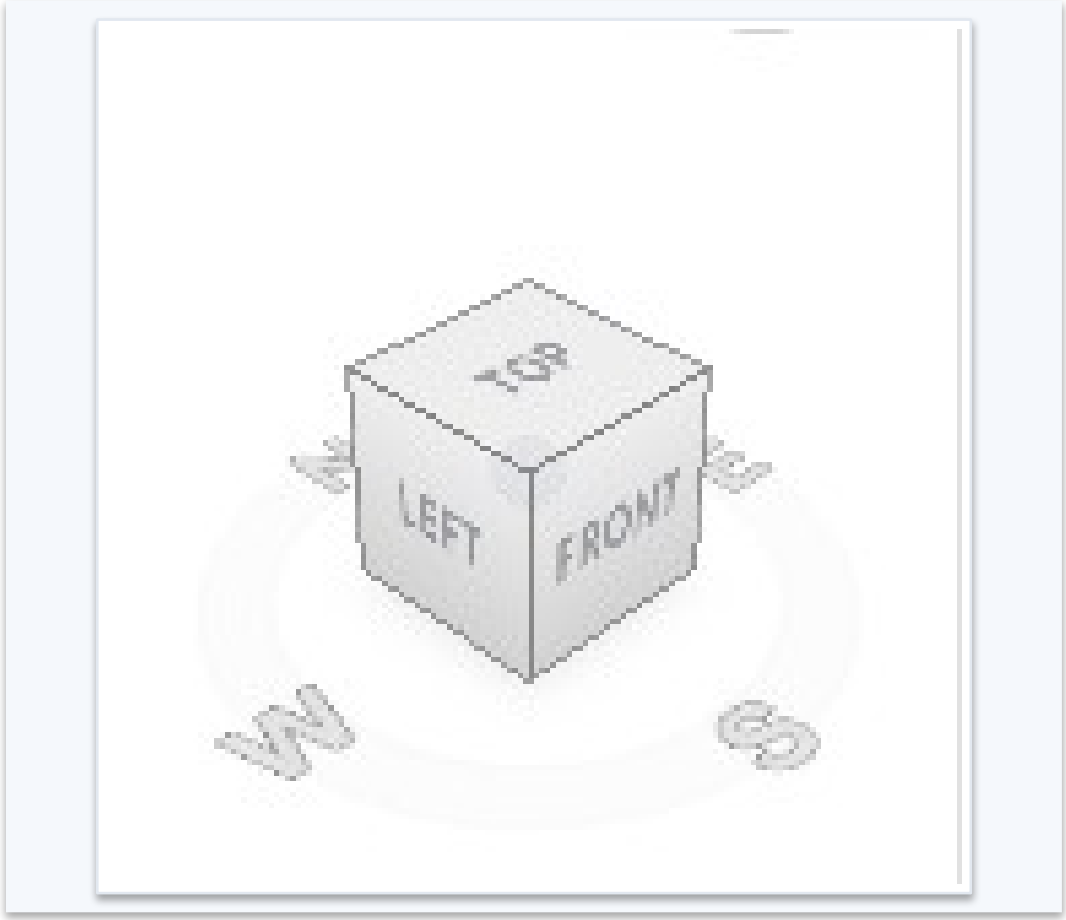


© 2026 Tertiary Infotech Academy Pte Ltd

NAVIGATION

ViewCube and 2D/3D Views

- The ViewCube switches between standard and isometric views of the model.
- Click a face, edge or corner to jump to that view; drag to orbit freely.
- Set Current View as Home to return to a saved orientation.
- In 2D drafting you will mostly stay on the Top view.

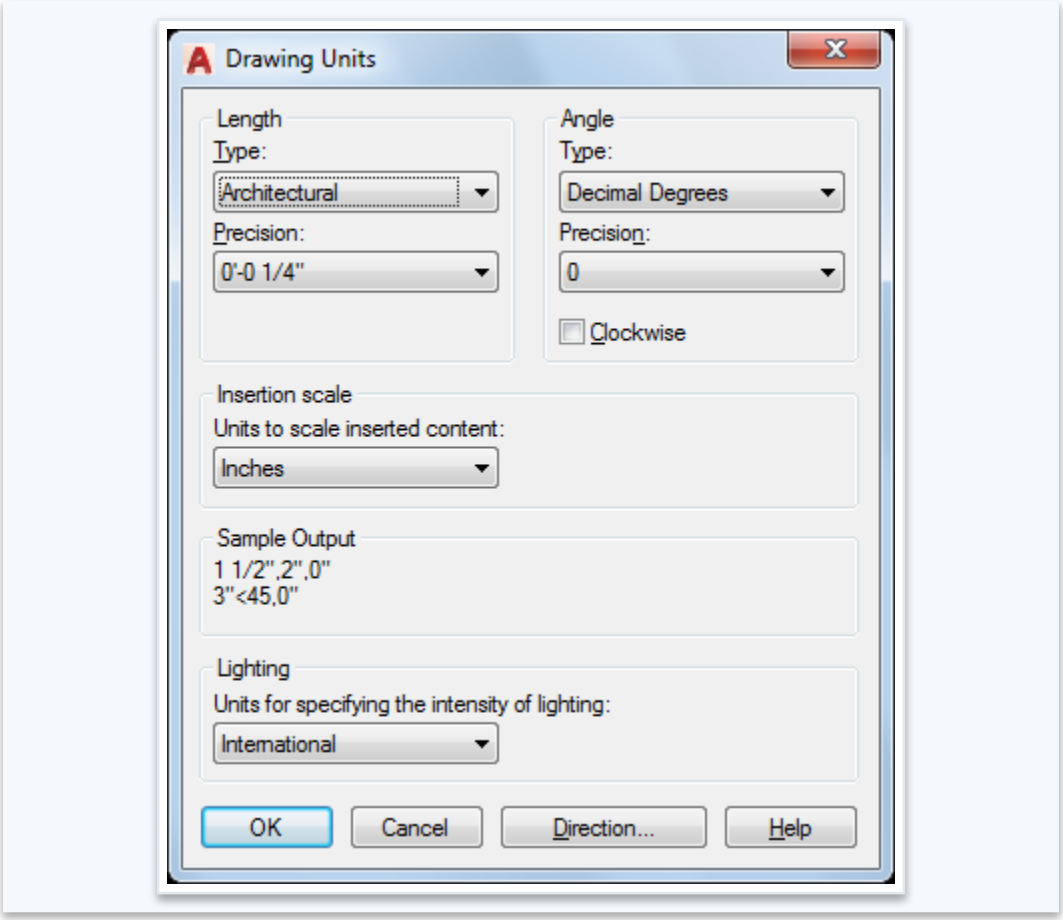


© 2026 Tertiary Infotech Academy Pte Ltd

SETUP

Drawing Units and Converting Units

- UNITS sets the length type (Decimal, Architectural, Engineering…) and precision.
- Angles default to decimal degrees measured anticlockwise from east.
- This course draws in millimetres with decimal units.
- To convert a drawing's scale: SCALE → All → base 0,0 → factor (e.g. 25.4).



© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 01 · HANDS-ON

Setting Up a New Drawing

LAB 1

Create a new metric drawing from the ACADISO template, choose the Drafting & Annotation workspace, set units and limits, then open a finished drawing and practise reading it with the zoom, pan and ViewCube navigation tools.

You'll produce

A correctly configured metric drawing (units, limits, workspace) saved as DWG, plus confident navigation of an existing technical drawing.

Commands: ACADISO template, UNITS, LIMITS, ZOOM, PAN, ViewCube
Files: labs/lab-01-setting-up-a-new-drawing/lab01-table.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 1 · STEP-BY-STEP

Setting Up a New Drawing

STEPS 1–3 OF 7

1	Start AutoCAD and create a new drawing from the ISO metric template	NEW → acadiso.dwt
2	Switch the workspace to Drafting & Annotation from the status bar gear icon	WSCURRENT → Drafting & Annotation
3	Set decimal length units with 0.00 precision and decimal degrees for angles	UNITS → Length: Decimal, Precision 0.00

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 1 · STEP-BY-STEP

Setting Up a New Drawing

STEPS 4–6 OF 7

4 Frame an A3-sized working area with drawing limits, then zoom to it

LIMITS → 0,0 → 420,297 | ZOOM → All

5 Toggle the precision aids on the status bar: grid, snap, ortho, polar and object snap

F7 grid · F9 snap · F8 ortho · F10 polar · F3 osnap

6 Open lab01-table.dwg and navigate it: wheel-zoom, pan, zoom window and zoom extents

OPEN lab01-table.dwg → Z → W / Z → E

© 2026 Tertiary Infotech Academy Pte Ltd

Setting Up a New Drawing

STEPS 7–7 OF 7

7

Identify the views, dimensions and title information in the drawing, then save your setup file

```
SAVEAS → lab01-setup.dwg
```


© 2026 Tertiary Infotech Academy Pte Ltd

LAB 1 · VERIFY

Setting Up a New Drawing

✅ Check your work

You can state the drawing's units, limits and workspace; zoom to any detail of lab01-table.dwg and back to extents; and your lab01-setup.dwg opens with the settings intact.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC RECAP

Recap — Getting Started with AutoCAD

1

Lab 1 — Setting Up a New Drawing

A correctly configured metric drawing (units, limits, workspace) saved as DWG, plus confident navigation of an existing technical drawing.

© 2026 Tertiary Infotech Academy Pte Ltd

02

TOPIC 02

Technical Drawing with AutoCAD

Basic AutoCAD Operations · Drawing Objects · Isometric Drawing

© 2026 Tertiary Infotech Academy Pte Ltd

SKILLS FRAMEWORK A2 · K2

Key Concepts — Technical Drawing with AutoCAD

1

Lines & polylines

LINE draws separate segments; PLINE creates one connected object of line and arc segments.

2

Circles & arcs

Circles by centre/radius or diameter; arcs by combinations of start, centre, end, radius and angle.

3

Rectangles & polygons

RECTANG builds closed rectangular polylines; POLYGON draws 3–1024-sided regular polygons.

4

Ellipses, splines & points

Ellipses from axis endpoints; splines as smooth curves through fit points; points as snap nodes.

5

Construction geometry

Construction lines (XLINE) and rays give infinite reference lines for projecting views.

6

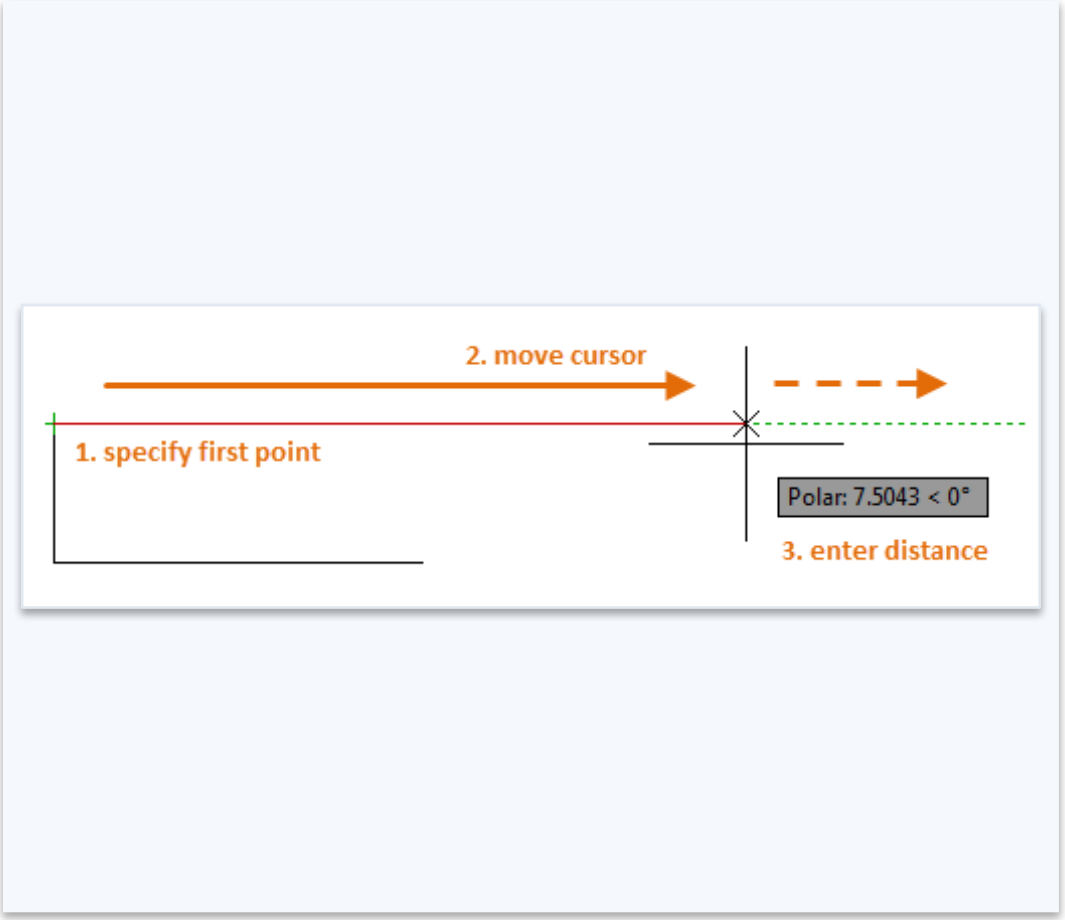
2D isometric drawing

ISODRAFT turns on isometric planes so a flat drawing represents a 3D projection.

© 2026 Tertiary Infotech Academy Pte Ltd

Lines

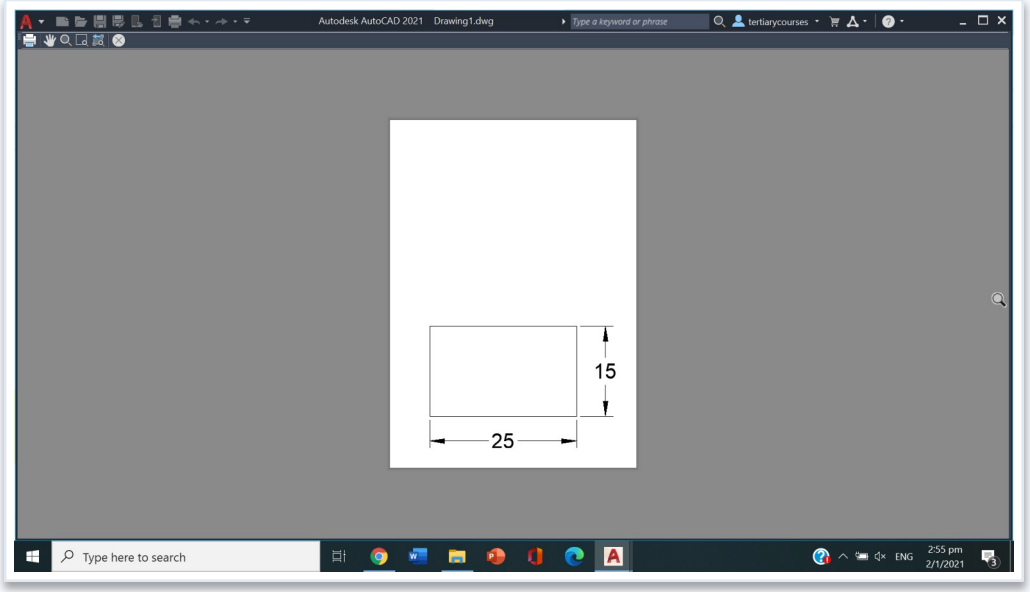
- LINE draws straight segments between picked or typed points.
- Specify the first point, then each next point; every segment is a separate object.
- Combine with object snaps and grid snap for exact geometry.
- Coordinate entry: absolute (100,50) · relative (@50,0) · polar (@50<30).



© 2026 Tertiary Infotech Academy Pte Ltd

Circles and Rectangles

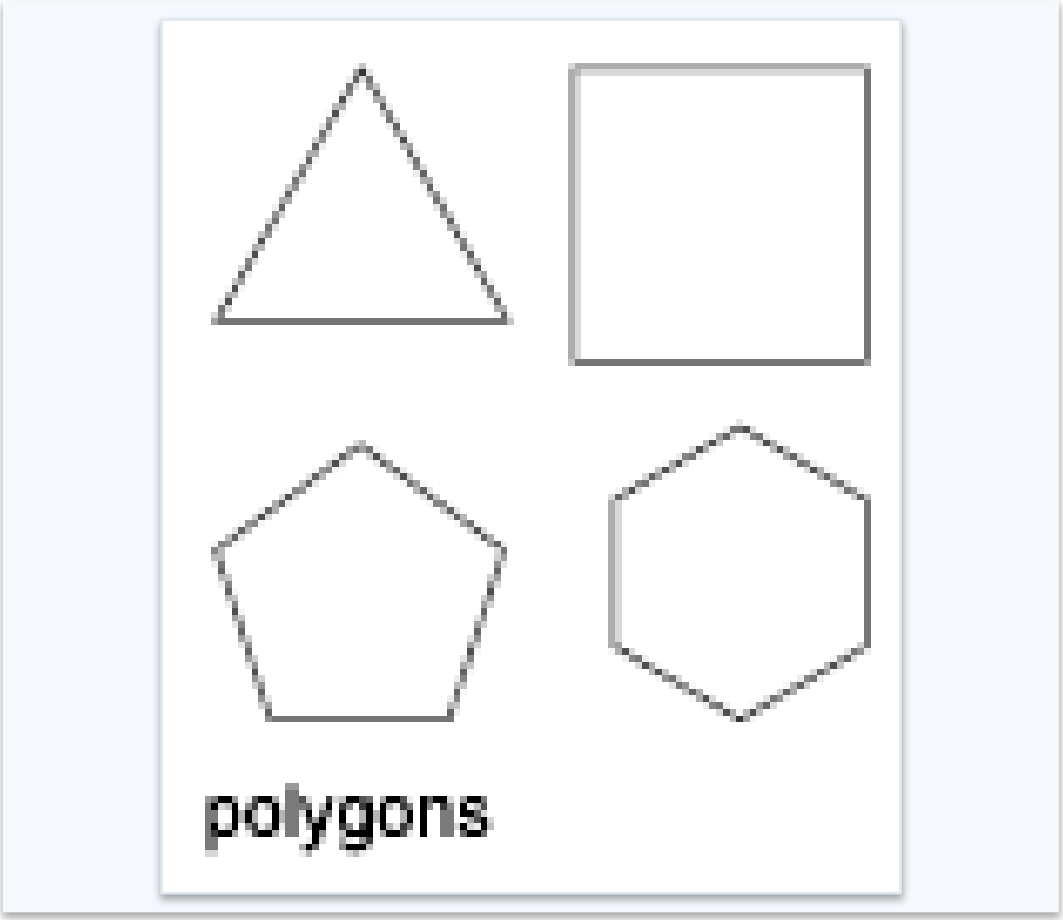
- CIRCLE: centre + radius or diameter; 2P/3P through points; TTR tangent-tangent-radius.
- RECTANG creates a closed rectangular polyline from two corners.
- Rectangle options add chamfered or filleted corners and a set width.
- Use dimensions option (@length,width) for exact rectangle sizes.



© 2026 Tertiary Infotech Academy Pte Ltd

Polygons and Ellipses

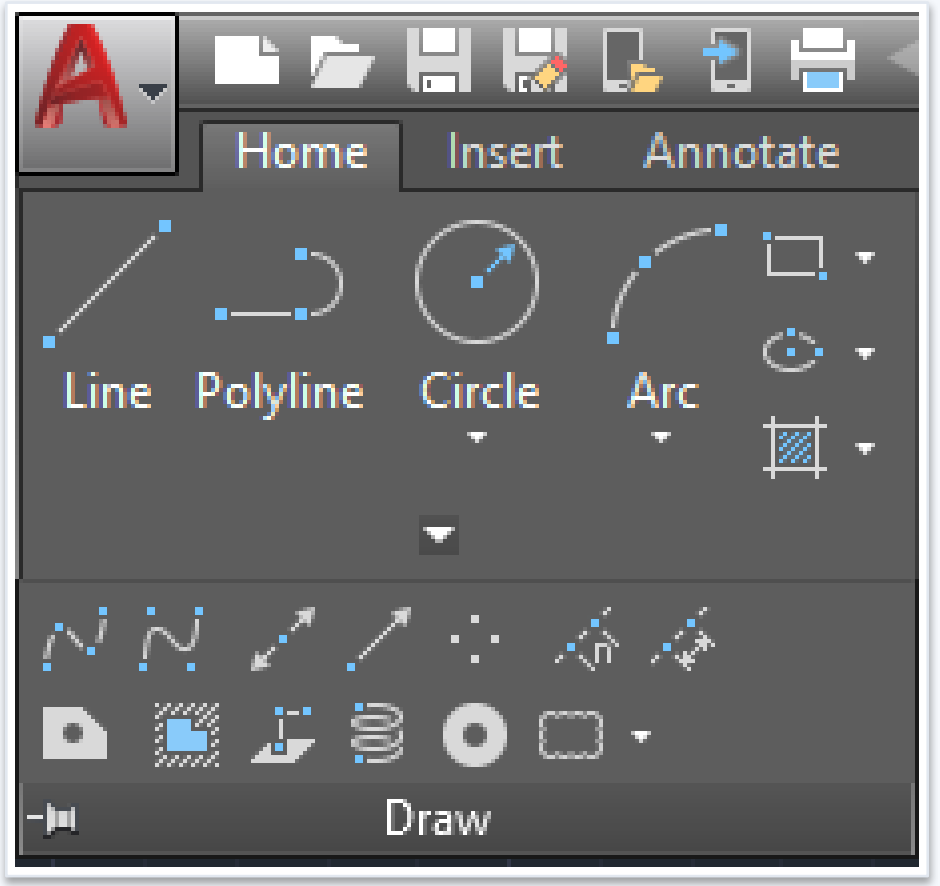
- POLYGON draws regular polygons of 3–1024 sides.
- Inscribed polygons fit inside the reference circle; circumscribed fit outside.
- ELLIPSE uses two axis endpoints then the half-width of the other axis.
- An elliptical arc is an ellipse with start and end angles.



© 2026 Tertiary Infotech Academy Pte Ltd

Arcs, Polylines and Splines

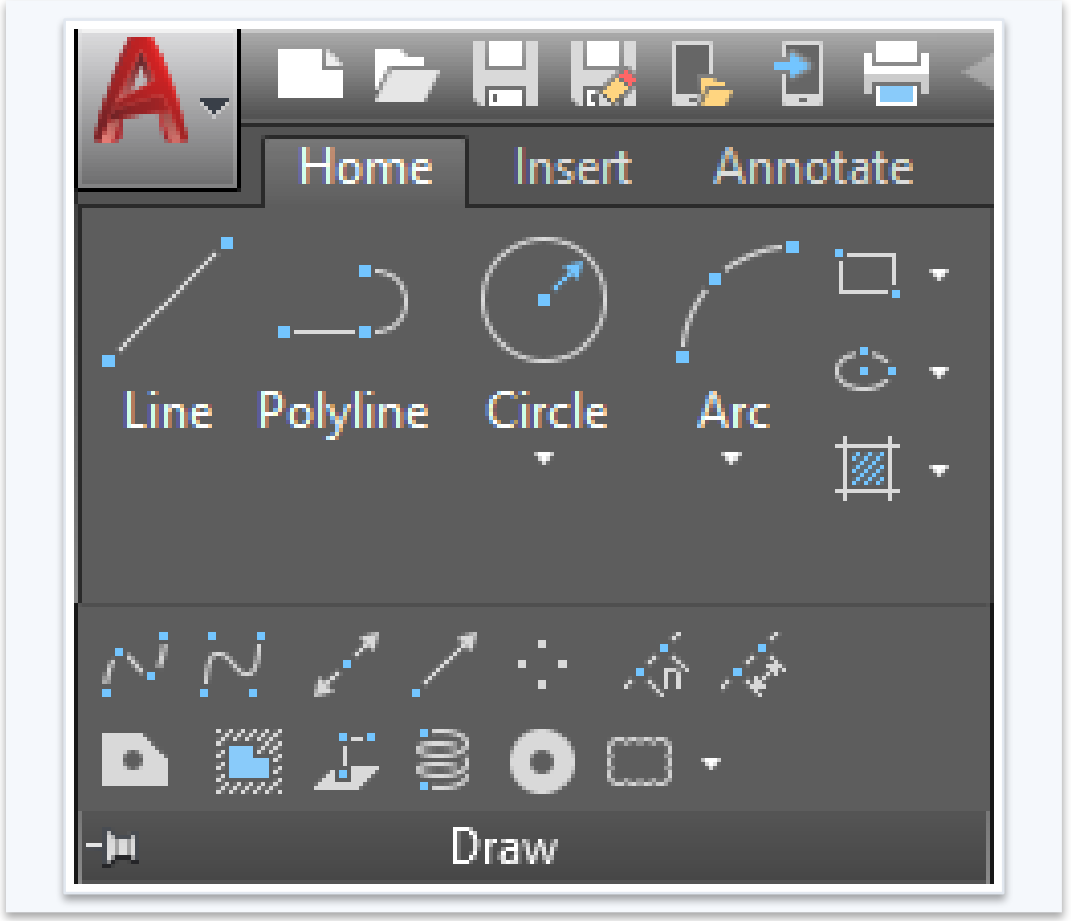
- ARC draws by combinations of start, centre, end, radius, angle and direction.
- PLINE creates one object of connected line and arc segments — switch modes as you draw.
- SPLINE fits a smooth curve through or near a set of fit points.
- Closed polylines are the basis for hatching, areas and 3D operations.



© 2026 Tertiary Infotech Academy Pte Ltd

Points, Construction Lines and Rays

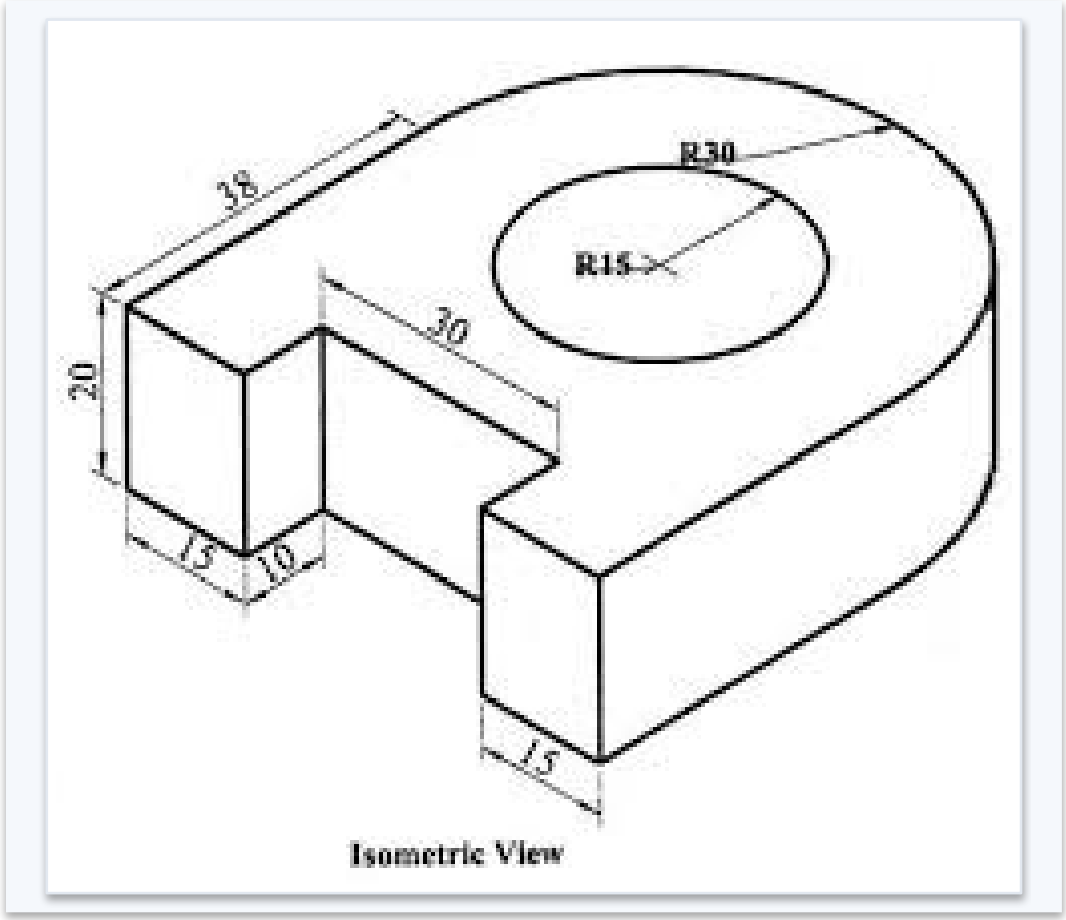
- POINT places node objects; PDMODE and PDSIZE control how they display.
- Points act as snap nodes — DIVIDE and MEASURE place them along objects.
- XLINE creates infinite construction lines for projecting between views.
- RAY starts at a point and extends infinitely in one direction.



© 2026 Tertiary Infotech Academy Pte Ltd

2D Isometric Drawing with ISODRAFT

- A 2D isometric drawing is a flat representation of a 3D isometric projection.
- ISODRAFT switches the crosshairs to the isometric axes (30°/90°/150°).
- F5 cycles the three isoplanes: Top, Right and Left.
- Isocircles (ELLIPSE → Isocircle) draw round features on the active isoplane.



© 2026 Tertiary Infotech Academy Pte Ltd

WHAT YOU'LL DO

Hands-On Labs — Technical Drawing with AutoCAD

Labs 2–3

- Creating Lines and Coordinates
- Circles, Rectangles and Polygons

Labs 4–5

- Polylines, Arcs, Ellipses and Splines
- 2D Isometric Drawing

—

- —

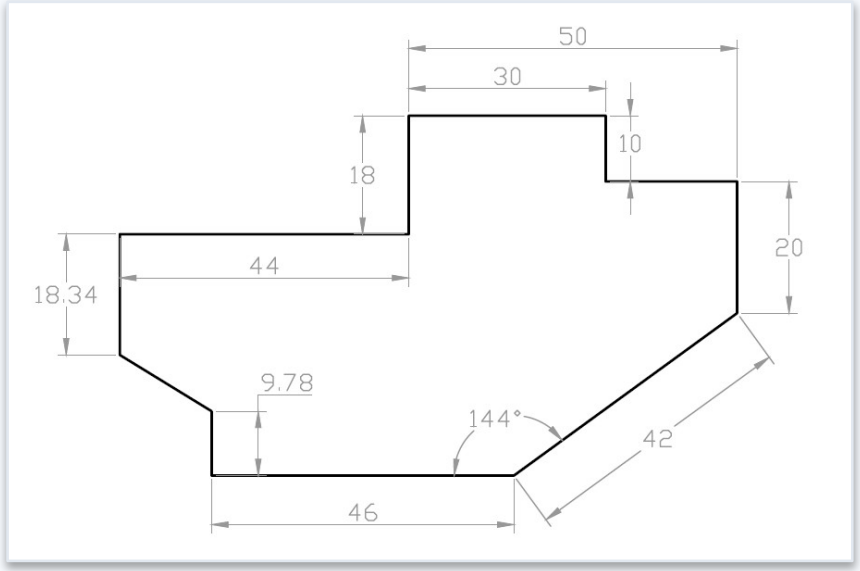
© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 02 · HANDS-ON

Creating Lines and Coordinates

LAB 2

Draw the outline of practice drawing EX-1 using the LINE and PLINE commands with absolute, relative and polar coordinate entry, working from its dimensioned drawing sheet.



You'll produce

The completed EX-1 profile drawn to size, matching the supplied solution DWG.

Commands: LINE, PLINE, coordinate entry (@dx,dy · @dist<angle), ORTHO, dynamic input

Files: labs/lab-02-creating-lines-and-coordinates/ex1.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 2 · STEP-BY-STEP

Creating Lines and Coordinates

STEPS 1–3 OF 6

1. Open the EX-1 drawing sheet (ex1-drawing-sheet.pdf) and study the dimensions

2. Start a new acadiso drawing and draw the first edge with absolute coordinates

```
L → 0,0 → 100,0
```

3. Continue the profile with relative coordinates for each dimensioned edge

```
@0,50 · @-30,0 · @20<135
```


© 2026 Tertiary Infotech Academy Pte Ltd

LAB 2 · STEP-BY-STEP

Creating Lines and Coordinates

STEPS 4–6 OF 6

4 Use ortho (F8) and polar tracking (F10) for the horizontal and vertical segments

F8 · F10

5 Close the profile back to the start point and check it forms one closed loop

C (Close)

6 Compare your result against the solution file ex1.dwg, then save

OPEN ex1.dwg → SAVEAS ex1-mine.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 2 · VERIFY

Creating Lines and Coordinates

✅ Check your work

Your outline closes exactly (LIST shows the final vertex at 0,0) and overlays the solution ex1.dwg with no visible deviation.

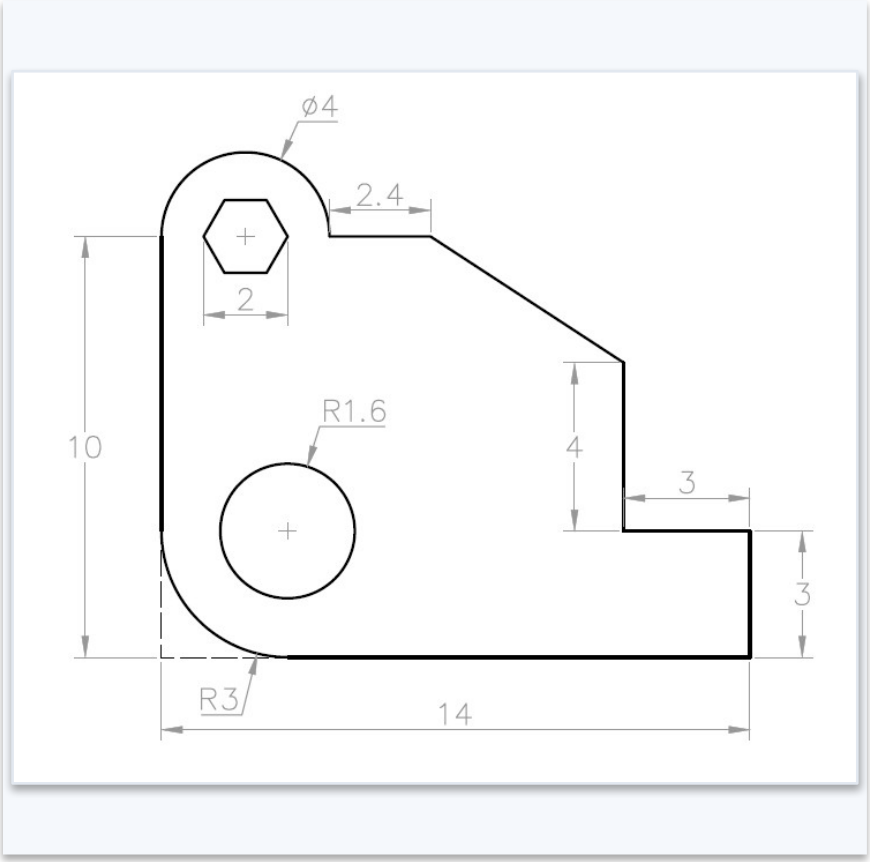
© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 02 · HANDS-ON

Circles, Rectangles and Polygons

LAB 3

Complete practice drawing EX-2 by adding circles (centre-radius, diameter, 2P/3P and Tan-Tan-Radius), a rectangle with chamfered/filleted corners and inscribed/circumscribed polygons.



You'll produce

The completed EX-2 drawing with all holes, rectangles and polygons placed to the sheet dimensions.

Commands: CIRCLE (Cen-Rad, 2P, 3P, TTR), RECTANG, POLYGON, object snaps
Files: labs/lab-03-circles-rectangles-and-polygons/ex2.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 3 · STEP-BY-STEP

Circles, Rectangles and Polygons

STEPS 1–3 OF 6

1 Open the EX-2 sheet (ex2-drawing-sheet.pdf) and identify each hole's centre

2 Draw the main circles by centre point and radius, snapping to intersections

C → cen point → 25

3 Add a circle by diameter and one through two points on existing geometry

C → D → 40 · C → 2P

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 3 · STEP-BY-STEP

Circles, Rectangles and Polygons

STEPS 4–6 OF 6

4 Draw a tangent-tangent-radius circle touching two existing circles

```
C → TTR → pick 2 tangents → radius
```

5 Create the rectangle feature, then a hexagon inscribed in a 30-radius circle

```
REC → corner → @80,50 · POL → 6 → I → 30
```

6 Verify against the solution ex2.dwg and save your drawing

```
SAVEAS ex2-mine.dwg
```


© 2026 Tertiary Infotech Academy Pte Ltd

LAB 3 · VERIFY

Circles, Rectangles and Polygons

✅ Check your work

Every circle's centre and radius match the sheet (use LIST), the hexagon has 6 equal sides, and your drawing overlays ex2.dwg cleanly.

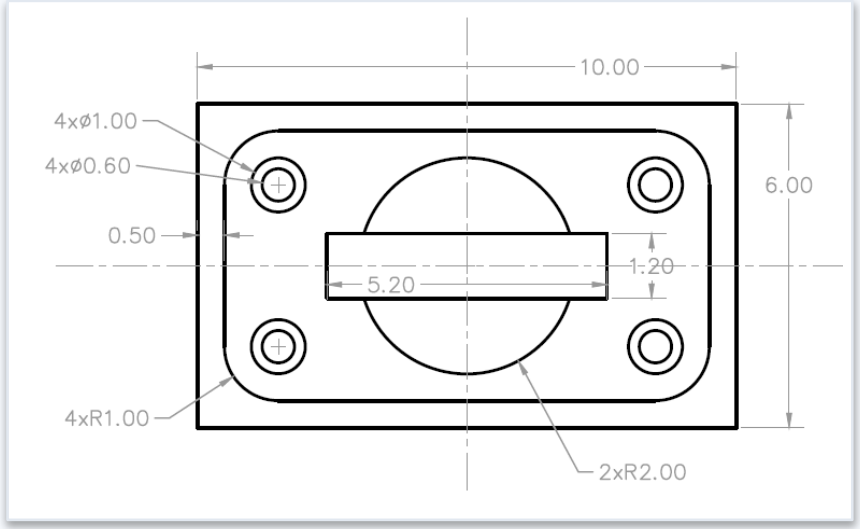
© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 02 · HANDS-ON

Polylines, Arcs, Ellipses and Splines

LAB 4

Draw practice drawing EX-3 combining polylines with arc segments, standalone arcs, an ellipse and a spline, plus construction lines to project feature positions between views.



You'll produce

The completed EX-3 drawing with mixed line/arc geometry drawn as clean, connected objects.

Commands: PLINE (Arc mode), ARC, ELLIPSE, SPLINE, XLINE, POINT/DIVIDE
Files: labs/lab-04-polylines-arcs-ellipses-and-splines/ex3.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 4 · STEP-BY-STEP

Polylines, Arcs, Ellipses and Splines

STEPS 1–3 OF 6

1 Open the EX-3 sheet (ex3-drawing-sheet.pdf) and plan the outline as one polyline

2 Draw the outline with PLINE, switching between Line and Arc segments as you go

PL → pick → A (arc) → L (line)

3 Place construction lines to project the centres of features between views

XL → H / V through centres

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 4 · STEP-BY-STEP

Polylines, Arcs, Ellipses and Splines

STEPS 4–6 OF 6

4 Add a three-point arc and a centre-start-end arc where the sheet shows open curves

A → 3 points · A → C

5 Draw the ellipse from its axis endpoints and a smooth spline through the fit points

EL → axis ends → half-width · SPL

6 Compare with ex3.dwg, purge the construction lines and save

ERASE xlines → SAVEAS ex3-mine.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 4 · VERIFY

Polylines, Arcs, Ellipses and Splines

✅ Check your work

The outline is a single polyline (one object when selected), arcs and the ellipse match the sheet dimensions, and no construction lines remain in the final drawing.

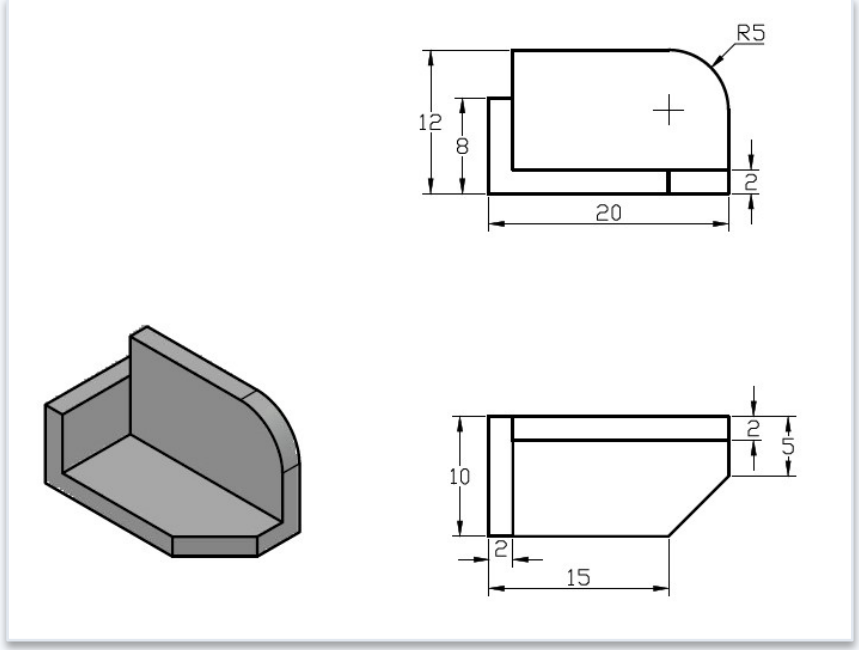
© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 02 · HANDS-ON

2D Isometric Drawing

LAB 5

Use ISODRAFT to create a 2D isometric representation of the 3D object supplied in iso-object.dwg, switching isoplanes and drawing isocircles for the round features.



You'll produce

A flat isometric drawing of the 3D object — edges on the three isoplanes plus isocircle holes.

Commands: ISODRAFT, isoplanes (F5), ELLIPSE Isocircle, polar tracking
Files: labs/lab-05-2d-isometric-drawing/iso-object.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 5 · STEP-BY-STEP

2D Isometric Drawing

STEPS 1–3 OF 6

1 Open the object sheet iso-object-sheet.pdf and identify its three faces

2 Turn on isometric drafting from the status bar and set the Top isoplane

ISODRAFT → Isoplane Top (F5)

3 Draw the top face edges along the 30°/150° isometric axes with polar tracking

L with polar 30°/150°

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 5 · STEP-BY-STEP

2D Isometric Drawing

STEPS 4–6 OF 6

4 Switch to the Right and Left isoplanes and complete the vertical faces

```
F5 → Right · F5 → Left
```

5 Add the round features as isocircles on the correct isoplane

```
EL → I (Isocircle) → centre → radius
```

6 Compare your isometric with the 3D model in iso-object.dwg and save

```
SAVEAS iso-mine.dwg
```


© 2026 Tertiary Infotech Academy Pte Ltd

LAB 5 · VERIFY

2D Isometric Drawing

✅ Check your work

Each face lies on the correct isoplane (edges at 30°/90°/150°), the isocircles read as ellipses aligned to their faces, and the drawing matches the supplied object's proportions.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC RECAP

Recap — Technical Drawing with AutoCAD

1

Lab 2 — Creating Lines and Coordinates

The completed EX-1 profile drawn to size, matching the supplied solution DWG.

2

Lab 3 — Circles, Rectangles and Polygons

The completed EX-2 drawing with all holes, rectangles and polygons placed to the sheet dimensions.

3

Lab 4 — Polylines, Arcs, Ellipses and Splines

The completed EX-3 drawing with mixed line/arc geometry drawn as clean, connected objects.

4

Lab 5 — 2D Isometric Drawing

A flat isometric drawing of the 3D object — edges on the three isoplanes plus isocircle holes.

© 2026 Tertiary Infotech Academy Pte Ltd

03

TOPIC 03

Modifying Objects

Move · Copy · Rotate · Offset · Mirror · Trim · Fillet · Arrays

© 2026 Tertiary Infotech Academy Pte Ltd

SKILLS FRAMEWORK A3 · K3

Key Concepts — Modifying Objects

1

Relocate & duplicate

MOVE, COPY and ROTATE reposition objects from a base point to a second point or angle.

2

Offset & mirror

OFFSET creates parallel copies at a set distance; MIRROR reflects half a symmetric design.

3

Trim & extend

Quick and Standard modes cut objects back to, or extend them to meet, boundary edges.

4

Fillet & chamfer

FILLET rounds a corner with a radius; CHAMFER bevels it with distances or an angle.

5

Arrays

Rectangular, polar and path arrays distribute copies in rows/columns, around a centre, or along a curve.

6

Stretch, scale & explode

STRETCH lengthens with a crossing window; SCALE resizes about a base point; EXPLODE breaks compounds.

© 2026 Tertiary Infotech Academy Pte Ltd

MODIFY

Relocating and Duplicating Objects

1

MOVE (M)

Select objects, pick a base point, then the second point — the objects relocate exactly.

2

COPY (CO)

Same flow as MOVE but leaves the original; Multiple mode places copy after copy.

3

ROTATE (RO)

Rotate about a base point by an absolute angle, or use Reference to match geometry.

4

OFFSET (O)

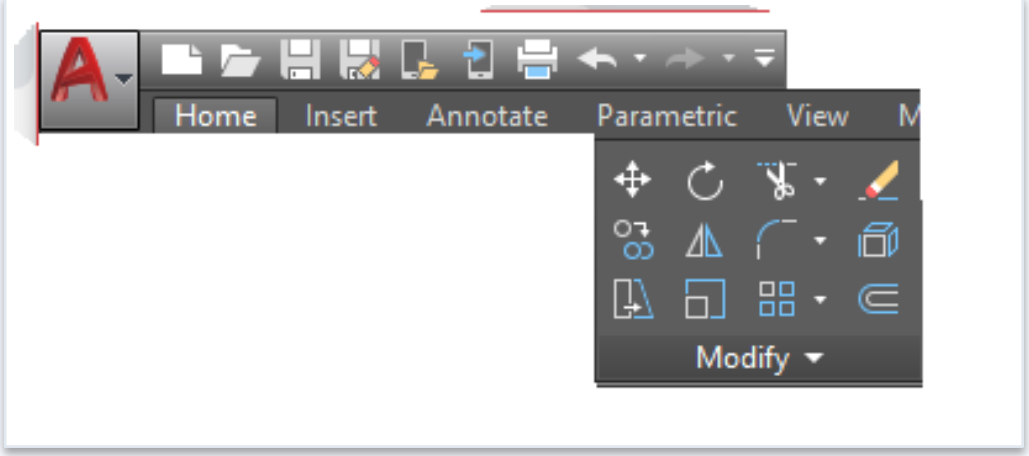
Creates concentric circles, parallel lines and parallel curves at a set distance.

© 2026 Tertiary Infotech Academy Pte Ltd

MODIFY

Offset in Practice

- Specify the offset distance, select the object, then click the side to offset to.
- Through mode offsets through a picked point instead of a distance.
- Offset polylines keep their corners; circles offset concentrically.
- Wall outlines, slots and rims are all one offset away from their centreline.



© 2026 Tertiary Infotech Academy Pte Ltd

MODIFY

Mirror, Stretch and Scale

1

MIRROR (MI)

Draw half of a symmetric design, then reflect it across a two-point mirror line.

2

STRETCH (S)

Select with a crossing window; endpoints inside move, the rest stay anchored.

3

SCALE (SC)

Resize about a base point by factor — or Reference to hit an exact length.

4

Grips

Select first, then drag grips to stretch/move without any command.

© 2026 Tertiary Infotech Academy Pte Ltd

MODIFY

Trim and Extend — Two Modes

Quick mode (default)

- Select the objects to be trimmed or extended, one by one or by dragging
- All visible geometry acts as the cutting/boundary edge
- Fastest for everyday cleanup

Standard mode

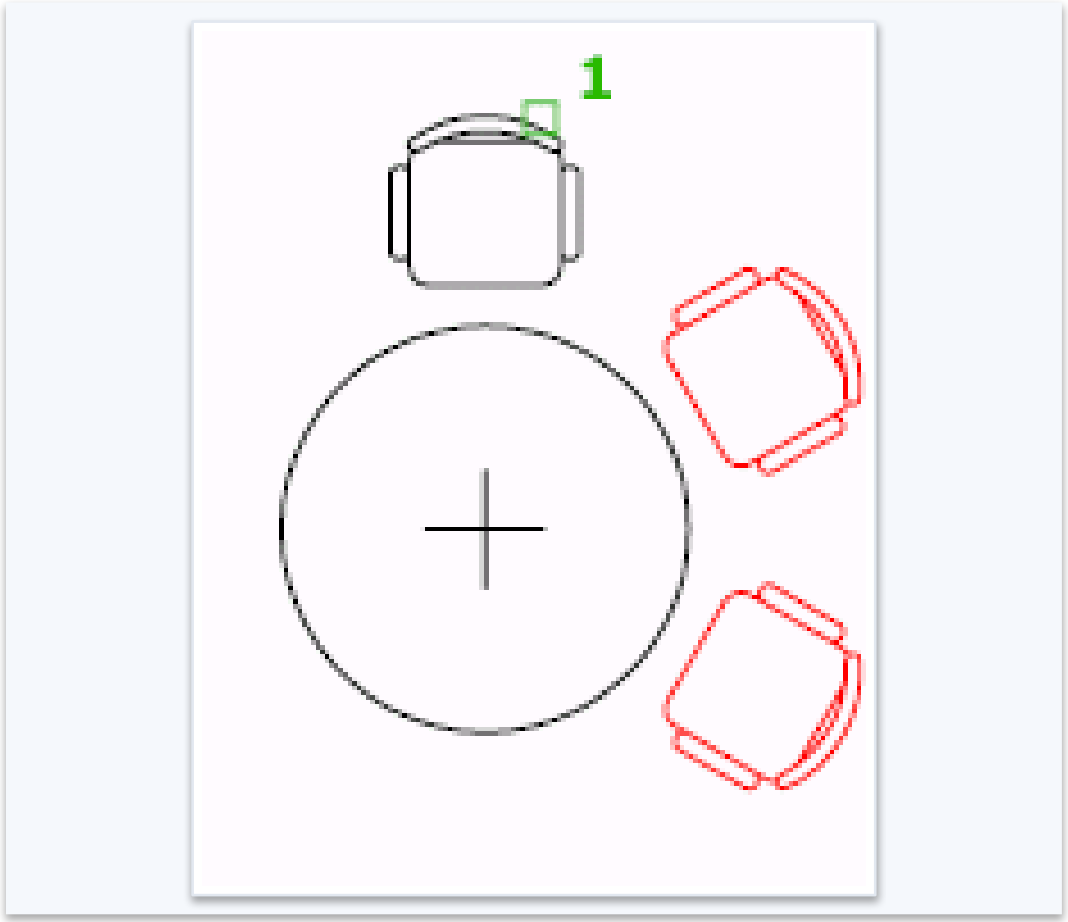
- First select boundary edges, press Enter, then select objects to trim/extend
- Press Enter at the first prompt to make everything a boundary
- Precise control for crowded drawings

© 2026 Tertiary Infotech Academy Pte Ltd

MODIFY

Arrays — Rectangular, Polar and Path

- ARRAY creates copies arranged in a pattern — associative by default.
- Rectangular: rows, columns and levels with set spacing.
- Polar: evenly distributes copies around a centre point or axis.
- Path: distributes copies along a line, polyline or curve.

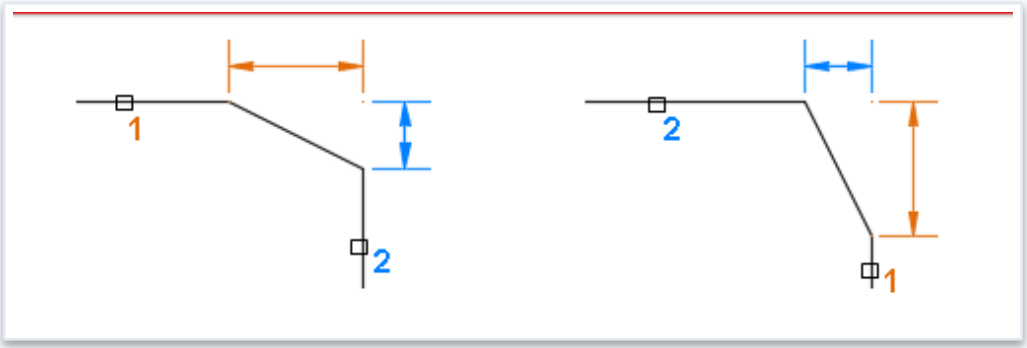


© 2026 Tertiary Infotech Academy Pte Ltd

MODIFY

Fillet and Chamfer

- FILLET rounds the corner between two objects with a set radius.
- Enter r (Radius) first to change the current fillet radius.
- CHAMFER bevels a corner using two distances or a distance and angle.
- Radius 0 / distance 0 are quick ways to square off a corner exactly.



© 2026 Tertiary Infotech Academy Pte Ltd

MODIFY

Erase, Explode, Break, Divide and Measure

1

ERASE (E)

Remove selected objects — or select first and press Delete.

2

EXPLODE (X)

Break a compound object (polyline, block, array) into components.

3

BREAK / Break at Point

Split lines, open polylines and arcs at one or two points.

4

DIVIDE · MEASURE

Place point nodes at equal counts (DIVIDE) or set intervals (MEASURE).

© 2026 Tertiary Infotech Academy Pte Ltd

WHAT YOU'LL DO

Hands-On Labs — Modifying Objects

Lab 6

- Move, Copy, Rotate and Offset

Lab 7

- Trim, Extend, Fillet and Chamfer

Lab 8

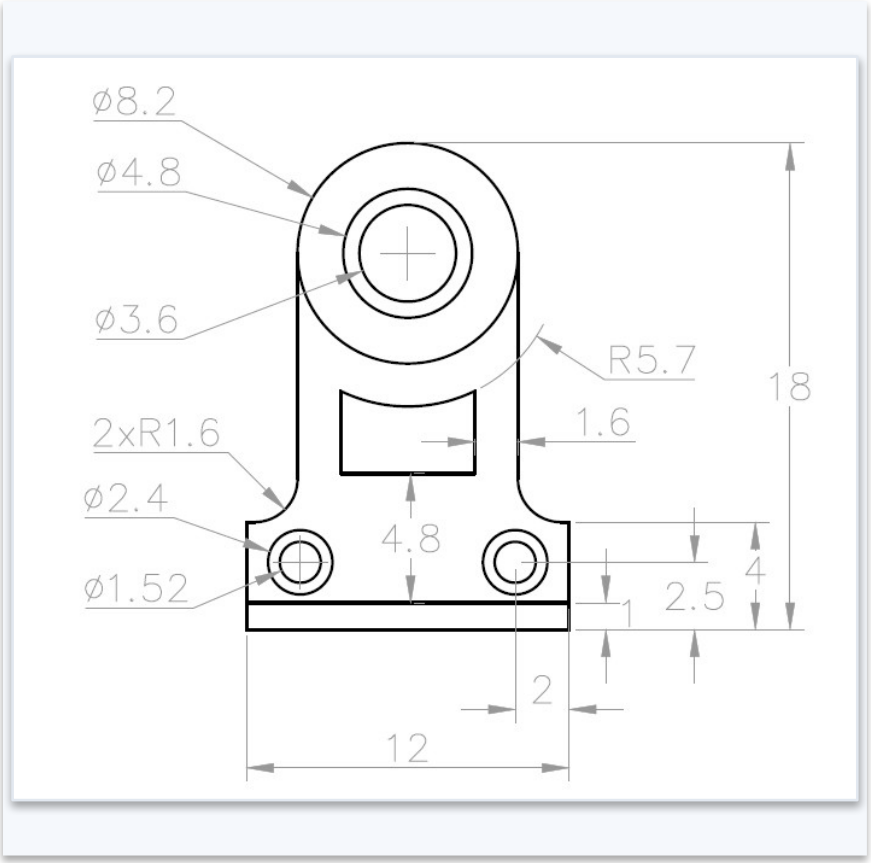
- Mirror, Stretch, Scale and Arrays

© 2026 Tertiary Infotech Academy Pte Ltd

Move, Copy, Rotate and Offset

LAB 6

Build practice drawing EX-4 by drawing one feature and then positioning the rest with MOVE, COPY (multiple), ROTATE with reference angle and OFFSET for the parallel profiles.



You'll produce

The completed EX-4 drawing produced with modify commands rather than re-drawing each feature.

Commands: MOVE, COPY (Multiple/Array), ROTATE (Reference), OFFSET
Files: labs/lab-06-move-copy-rotate-and-offset/ex4.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 6 · STEP-BY-STEP

Move, Copy, Rotate and Offset

STEPS 1–3 OF 6

1 Open the EX-4 sheet (ex4-drawing-sheet.pdf) and spot the repeated features

2 Draw the base profile, then offset it to create the parallel inner/outer edges

O → distance → pick side

3 Draw one hole feature and copy it to every dimensioned centre

CO → base point → second points

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 6 · STEP-BY-STEP

Move, Copy, Rotate and Offset

STEPS 4–6 OF 6

4 Move a mis-placed feature precisely from its centre to the correct intersection

```
M → base → destination
```

5 Rotate the angled feature about its base point, using Reference to match the sheet angle

```
RO → base → R → ref angle → new angle
```

6 Check against ex4.dwg and save your version

```
SAVEAS ex4-mine.dwg
```


© 2026 Tertiary Infotech Academy Pte Ltd

LAB 6 · VERIFY

Move, Copy, Rotate and Offset

✅ Check your work

Feature spacings measure correct with DIST, offsets are at the specified distances, and the rotated feature sits at exactly the sheet's angle.

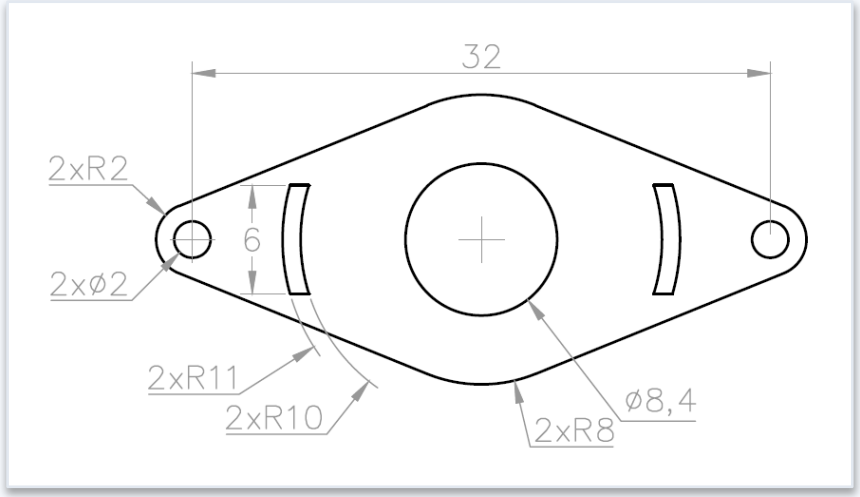
© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 03 · HANDS-ON

Trim, Extend, Fillet and Chamfer

LAB 7

Complete practice drawing EX-5 by trimming overlapping construction geometry, extending short edges to their boundaries, and finishing corners with fillets and chamfers.



You'll produce

A clean EX-5 profile with every corner finished exactly as dimensioned on the sheet.

Commands: TRIM (Quick/Standard), EXTEND, FILLET, CHAMFER, BREAK
Files: labs/lab-07-trim-extend-fillet-and-chamfer/ex5.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 7 · STEP-BY-STEP

Trim, Extend, Fillet and Chamfer

STEPS 1–3 OF 6

1 Open the EX-5 sheet (ex5-drawing-sheet.pdf) and rough out the profile with overlapping lines

2 Trim the overshoots in Quick mode — click or drag across the pieces to remove

```
TR → click segments
```

3 Use Standard mode where you need explicit cutting edges: select boundaries, then objects

```
TR → Enter (all edges) → pick
```


© 2026 Tertiary Infotech Academy Pte Ltd

LAB 7 · STEP-BY-STEP

Trim, Extend, Fillet and Chamfer

STEPS 4–6 OF 6

4 Extend the short edges to meet their boundary lines

```
EX → pick edges to extend
```

5 Fillet the rounded corners at the sheet radius, and chamfer the bevelled ones

```
F → R → 10 → pick 2 lines · CHA → D → 5,5
```

6 Compare with ex5.dwg and save

```
SAVEAS ex5-mine.dwg
```


© 2026 Tertiary Infotech Academy Pte Ltd

LAB 7 · VERIFY

Trim, Extend, Fillet and Chamfer

✅ Check your work

No overshooting or dangling segments remain (zoom around the whole profile), and every fillet radius / chamfer distance matches the drawing sheet.

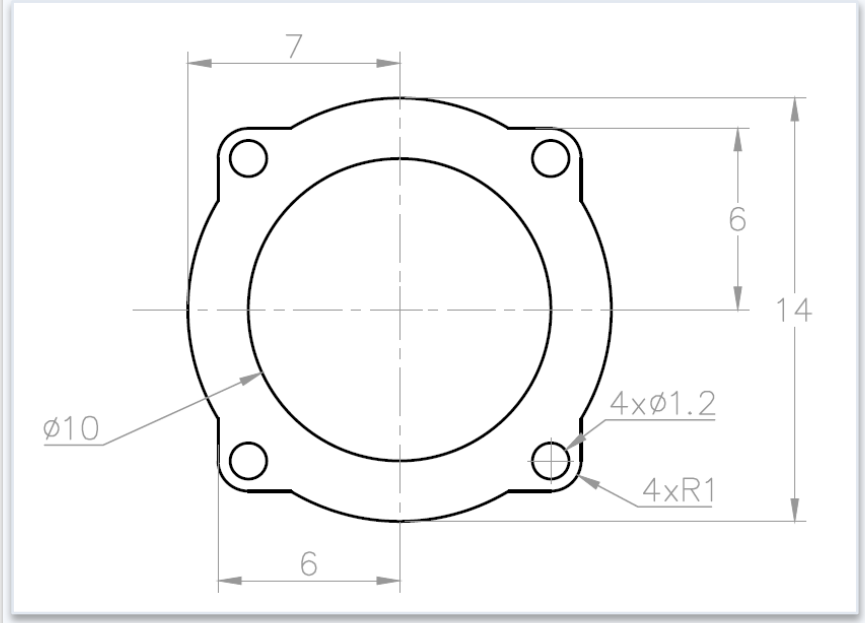
© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 03 · HANDS-ON

Mirror, Stretch, Scale and Arrays

LAB 8

Produce practice drawing EX-6 by drawing half of the symmetric design and mirroring it, then distributing the hole pattern with rectangular and polar arrays, and adjusting sizes with STRETCH and SCALE.



You'll produce

The completed EX-6 drawing with its symmetric outline and evenly arrayed hole pattern.

Commands: MIRROR, STRETCH (crossing window), SCALE (Reference), ARRAYRECT, ARRAYPOLAR

Files: labs/lab-08-mirror-stretch-scale-and-arrays/ex6.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 8 · STEP-BY-STEP

Mirror, Stretch, Scale and Arrays

STEPS 1–3 OF 6

1 Open the EX-6 sheet (ex6-drawing-sheet.pdf) and find its axis of symmetry

2 Draw one half of the outline, then mirror it across the centreline

MI → select → axis pt1 → pt2 → N

3 Array one hole into the rectangular pattern (rows, columns, spacing from the sheet)

AR → R → rows/cols/spacing

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 8 · STEP-BY-STEP

Mirror, Stretch, Scale and Arrays

STEPS 4–6 OF 6

4 Distribute the bolt holes around the centre with a polar array

```
AR → PO → centre → items → 360
```

5 Stretch the slot to its dimensioned length with a crossing window, scale a feature with Reference

```
S → crossing → base → distance · SC → R
```

6 Verify against ex6.dwg and save

```
SAVEAS ex6-mine.dwg
```


© 2026 Tertiary Infotech Academy Pte Ltd

LAB 8 · VERIFY

Mirror, Stretch, Scale and Arrays

✅ Check your work

The two halves are perfectly symmetric about the centreline, array counts and spacings match the sheet, and the stretched slot measures the dimensioned length.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC RECAP

Recap — Modifying Objects

1

Lab 6 — Move, Copy, Rotate and Offset

The completed EX-4 drawing produced with modify commands rather than re-drawing each feature.

2

Lab 7 — Trim, Extend, Fillet and Chamfer

A clean EX-5 profile with every corner finished exactly as dimensioned on the sheet.

3

Lab 8 — Mirror, Stretch, Scale and Arrays

The completed EX-6 drawing with its symmetric outline and evenly arrayed hole pattern.

© 2026 Tertiary Infotech Academy Pte Ltd

04

TOPIC 04

Organizing Objects

Object Properties · Layers · Blocks, Attributes & Reusable Content

© 2026 Tertiary Infotech Academy Pte Ltd

Key Concepts — Organizing Objects

1

Object properties

Colour, linetype and lineweight — set ByLayer and inspected in the Properties palette (PR).

2

Layers

The primary method of organising a drawing by function; control on/off, freeze/thaw, lock and plot.

3

Grips

Multi-functional grips stretch, move, rotate, scale and copy objects without typing a command.

4

Hide & isolate

Temporarily isolate the objects you are working on and hide the rest of the drawing.

5

Blocks & attributes

Blocks are reusable symbols; attributes attach part numbers and data — the basis of part lists.

6

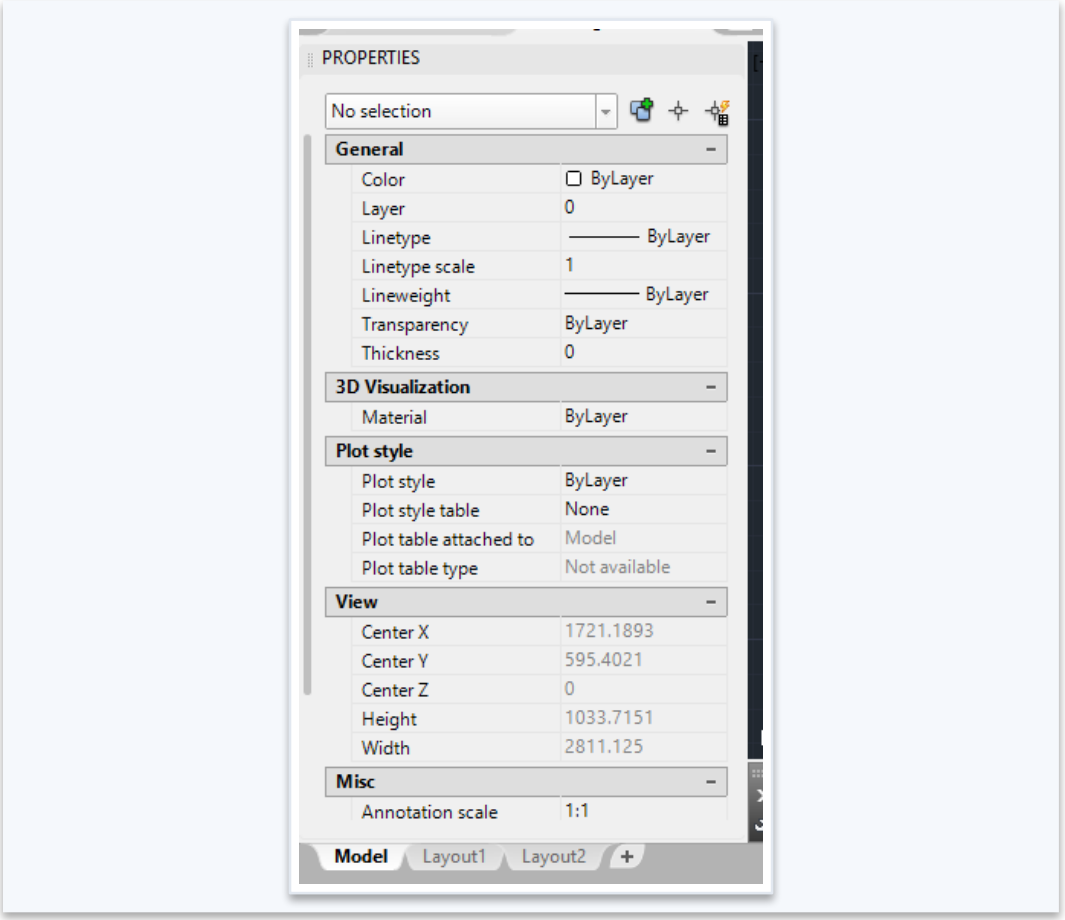
External references

Xrefs attach another drawing (or PDF underlay) so teams share one live reference file.

© 2026 Tertiary Infotech Academy Pte Ltd

Object Properties and the Properties Palette

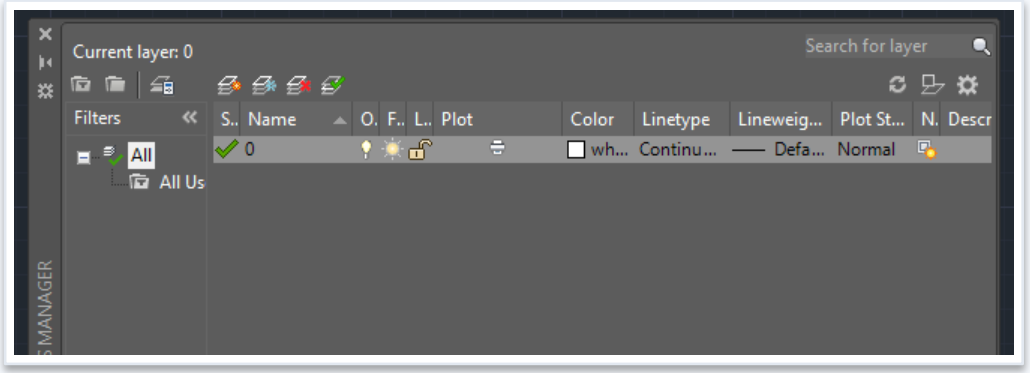
- The Home tab's Layers and Properties panels set the most common properties.
- The Properties palette (PR / Ctrl+1) is the complete list for any selection.
- With nothing selected it shows — and sets — the current defaults.
- Keep colour, linetype and lineweight ByLayer so layers stay in control.



© 2026 Tertiary Infotech Academy Pte Ltd

Layers — Organising Every Drawing

- Layers organise objects by function: outlines, hidden lines, centrelines, text, dimensions.
- The Layer Properties Manager creates layers and sets colour, linetype, lineweight and plot.
- All new objects are drawn on the current layer (green tick).
- Off vs freeze: frozen layers are skipped entirely and speed up big drawings; lock to view but not edit.

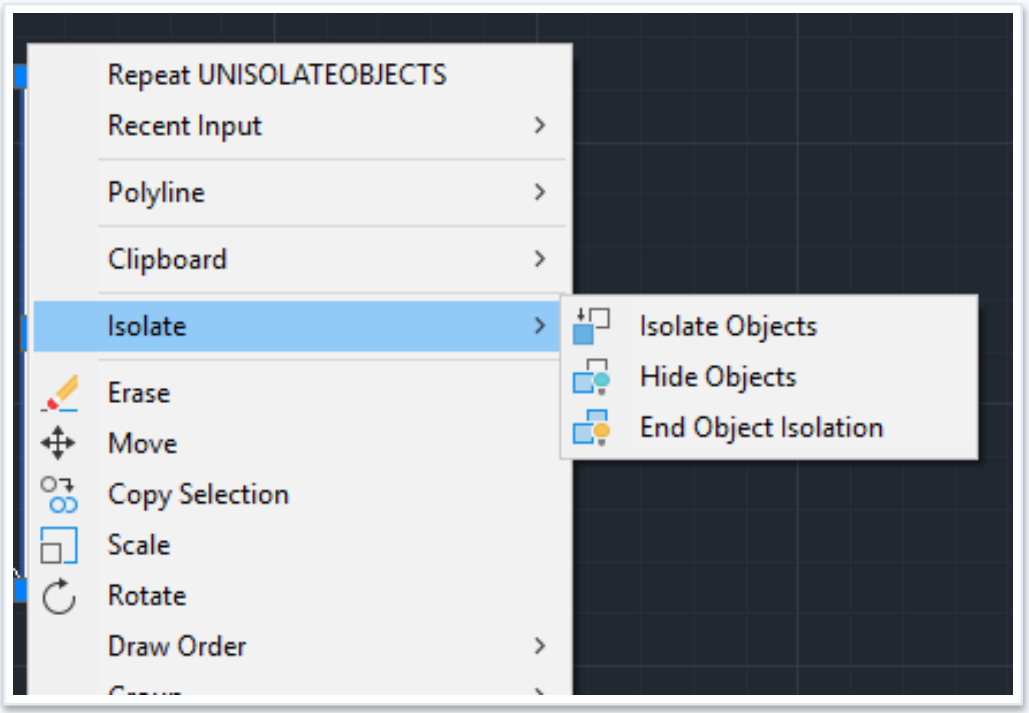


© 2026 Tertiary Infotech Academy Pte Ltd

FOCUS

Hide and Isolate Objects

- Isolate Selected Objects hides everything except your selection.
- Hide Selected Objects removes just the selection from view.
- Both are temporary view states — End Object Isolation restores all.
- Right-click menu or the status-bar lightbulb icon controls isolation.

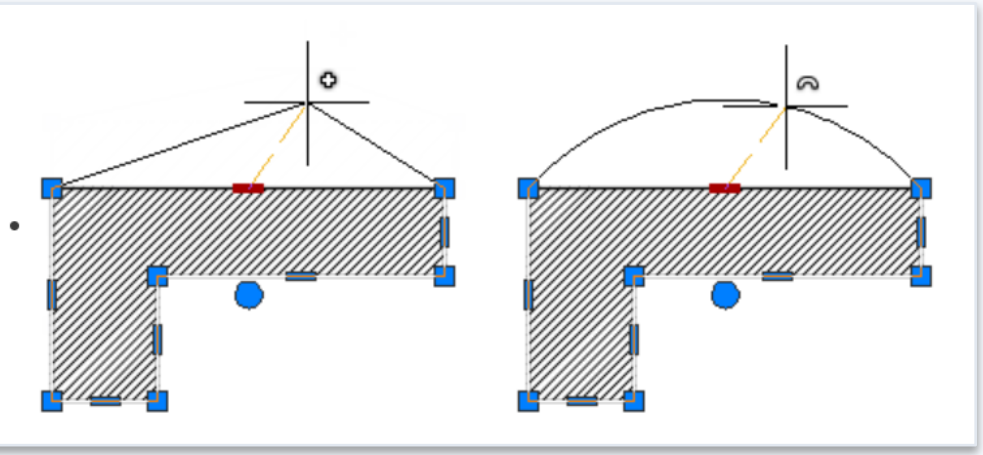


© 2026 Tertiary Infotech Academy Pte Ltd

DIRECT EDITING

Multi-Functional Grips

- Select with no command running and grips appear at defining points.
- Drag a grip to stretch; press Enter to cycle Stretch → Move → Rotate → Scale → Mirror.
- Hover a grip for its multi-function menu (e.g. add vertex, convert to arc).
- Lines, polylines, arcs, dimensions and viewports all expose smart grips.

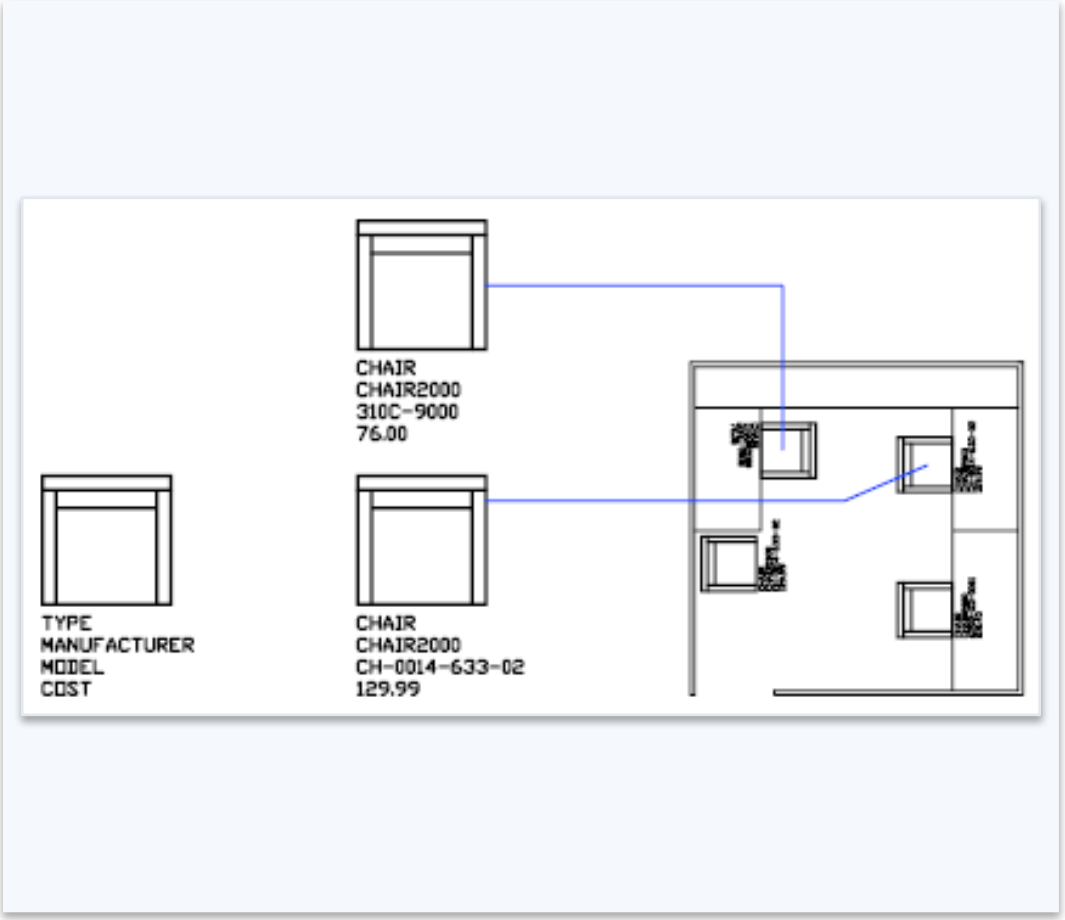


© 2026 Tertiary Infotech Academy Pte Ltd

REUSABLE CONTENT

Blocks, Dynamic Blocks and Attributes

- BLOCK combines objects into a named, reusable symbol inserted with INSERT.
- Dynamic blocks add parameters and actions — one block, many sizes/states.
- Attributes attach data to a block: part numbers, descriptions, quantities.
- ATTDEF defines the tag, prompt and default; DATAEXTRACTION turns attributes into part-list tables.

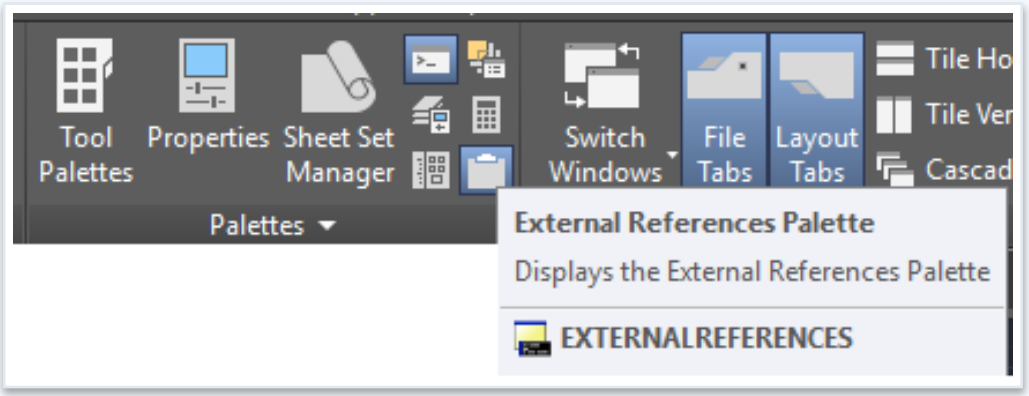


© 2026 Tertiary Infotech Academy Pte Ltd

REUSABLE CONTENT

External References (Xrefs)

- An xref attaches another drawing file into the current drawing — live, not copied.
- The External References palette organises, reloads and repaths every reference.
- Teams share one title block or site plan; edits flow to every host on reload.
- PDFs and images attach the same way as underlays for tracing.



© 2026 Tertiary Infotech Academy Pte Ltd

WHAT YOU'LL DO

Hands-On Labs — Organizing Objects

Labs 9–10

- Object Properties and Layers
- Grips, Hide and Isolate

Labs 11–12

- Blocks, Attributes and Part Lists
- External References and PDF Import

—

- —

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 04 · HANDS-ON

Object Properties and Layers

LAB 9

Open the furniture drawing lab-table-lesson2.dwg, create a layer scheme (outline, hidden, centre, dimensions, text), assign colours/linetypes/lineweights ByLayer, and move the existing objects onto the correct layers.

You'll produce

A layered version of the table drawing where every object's colour and linetype come ByLayer.

Commands: LAYER (LA), Properties palette (PR), Layer Walk, MATCHPROP, freeze/thaw, lock
Files: labs/lab-09-object-properties-and-layers/lab-table-lesson2.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 9 · STEP-BY-STEP

Object Properties and Layers

STEPS 1–3 OF 6

1	Open lab-table-lesson2.dwg and inspect current object properties	PR (Properties palette)
2	Create layers: A-OUTLINE, A-HIDDEN, A-CENTRE, A-DIMS, A-TEXT with colours and linetypes	LA → New Layer → colour/linetype
3	Load the HIDDEN and CENTER linetypes and set lineweights per layer	LA → Linetype → Load → HIDDEN, CENTER

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 9 · STEP-BY-STEP

Object Properties and Layers

STEPS 4–6 OF 6

4 Select objects by type and move them to their layers; copy formatting with MATCHPROP

`select → Layer drop-down · MA`

5 Practise layer states: turn off, freeze/thaw, lock and Layer Walk through the scheme

`LAYWALK · LAYFRZ · LAYLCK`

6 Set every object's colour, linetype and lineweight to ByLayer and save

`SETBYLAYER → SAVEAS table-layers.dwg`

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 9 · VERIFY

Object Properties and Layers

✅ Check your work

Turning each layer off hides exactly its objects, everything reports ByLayer in the Properties palette, and Layer Walk steps cleanly through the scheme.

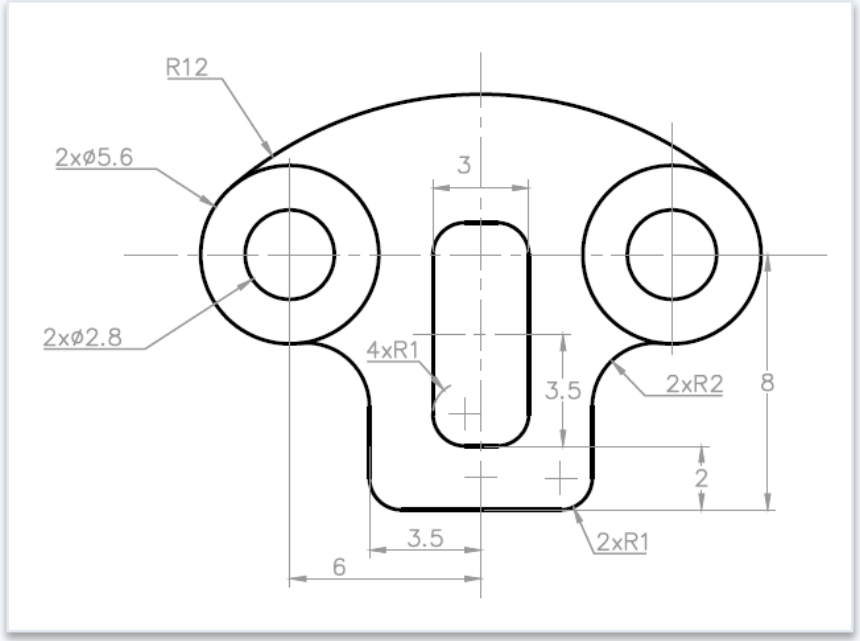
© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 04 · HANDS-ON

Grips, Hide and Isolate

LAB 10

Use multi-functional grips on practice drawing EX-7 to stretch, move, copy and rotate features without commands, and manage a busy view with Isolate and Hide Selected Objects.



You'll produce

An edited EX-7 layout adjusted entirely with grips, plus fluent use of isolate/hide/end-isolate.

Commands: Multi-functional grips, grip modes (Enter to cycle), Isolate/Hide Objects, Select Similar

Files: labs/lab-10-grips-hide-and-isolate/ex7.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 10 · STEP-BY-STEP

Grips, Hide and Isolate

STEPS 1–3 OF 6

1 Open ex7.dwg, click an object and hover its grips to see the grip menu

2 Stretch an endpoint grip to lengthen an edge; use the midpoint grip to move the object

```
click grip → drag / type distance
```

3 Cycle grip modes with Enter: Stretch → Move → Rotate → Scale → Mirror

```
select grip → Enter to cycle
```


© 2026 Tertiary Infotech Academy Pte Ltd

LAB 10 · STEP-BY-STEP

Grips, Hide and Isolate

STEPS 4–6 OF 6

4 Copy a feature by grip-moving with the Copy option (or Ctrl held)

```
grip → Move → C (Copy)
```

5 Right-click a feature: Isolate Objects → Isolate Selected, edit it, then End Isolation

```
ISOLATEOBJECTS · UNISOLATEOBJECTS
```

6 Use Select Similar to grab all matching objects and hide them temporarily

```
select → right-click → Select Similar → Hide
```


© 2026 Tertiary Infotech Academy Pte Ltd

LAB 10 · VERIFY

Grips, Hide and Isolate

✅ Check your work

You can stretch/move/rotate/copy with grips alone (no typed commands), and isolate then restore any subset of the drawing without losing objects.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 04 · HANDS-ON

Blocks, Attributes and Part Lists

LAB 11

Define a reusable part-tag block with attributes (part number, description, quantity) in the title-block drawing, insert it against each component, and prepare the attribute data that will feed the part list.

You'll produce

A BALLOON block with PART_NO / DESC / QTY attributes, inserted and filled in for each part.

Commands: BLOCK, ATTDEF, INSERT, ATTEDIT, groups, DATAEXTRACTION (preview)
Files: labs/lab-11-blocks-attributes-and-part-lists/title-block.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 11 · STEP-BY-STEP

Blocks, Attributes and Part Lists

STEPS 1–3 OF 6

1 Open title-block.dwg and draw the balloon symbol (circle + leader stub)

C → radius 5

2 Define attributes PART_NO, DESC and QTY with prompts and default values

ATTDEF → tag/prompt/default

3 Create the block BALLOON from the circle plus its attribute definitions

B → name BALLOON → select objects

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 11 · STEP-BY-STEP

Blocks, Attributes and Part Lists

STEPS 4–6 OF 6

4 Insert the balloon against each component and fill in the attribute values

I → BALLOON → answer prompts

5 Double-click a balloon to edit its attributes; group related parts together

ATTEDIT / EATTEDIT · GROUP

6 Preview extracting the attribute data — this becomes the part list in Lab 14

DATAEXTRACTION

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 11 · VERIFY

Blocks, Attributes and Part Lists

✅ Check your work

Each inserted balloon prompts for and displays its own part number, description and quantity, and Data Extraction lists one row per balloon with the values you entered.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 04 · HANDS-ON

External References and PDF Import

LAB 12

Attach the title-block drawing as an external reference under the table drawing, observe how xref updates flow through, then attach a PDF as an underlay and import PDF geometry.

You'll produce

A host drawing with a live xref title block and a PDF underlay ready for tracing.

Commands: XREF/ATTACH, External References palette, PDFATTACH, PDFIMPORT
Files: labs/lab-12-external-references-and-pdf-import/title-block.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 12 · STEP-BY-STEP

External References and PDF Import

STEPS 1–3 OF 6

1. Open your layered table drawing from Lab 9 as the host drawing

```
OPEN table-layers.dwg
```

2. Attach title-block.dwg as an Xref at 0,0, scale 1

```
XATTACH → title-block.dwg
```

3. Open the External References palette and note the xref's status and path

```
XR (External References)
```


© 2026 Tertiary Infotech Academy Pte Ltd

LAB 12 · STEP-BY-STEP

External References and PDF Import

STEPS 4–6 OF 6

4 Edit the title block in its own file, save, and reload the xref in the host

`XR → right-click → Reload`

5 Attach the supplied drawing-sheet PDF as an underlay and snap to its geometry

`PDFATTACH → ex1-drawing-sheet.pdf`

6 Import the PDF's vector geometry into the drawing and inspect the created objects

`PDFIMPORT → select underlay area`

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 12 · VERIFY

External References and PDF Import

✅ Check your work

Changing the title block file and reloading updates the host without copying geometry, and PDFIMPORT produces editable AutoCAD objects from the underlay.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC RECAP

Recap — Organizing Objects

1

Lab 9 — Object Properties and Layers

A layered version of the table drawing where every object's colour and linetype come ByLayer.

2

Lab 10 — Grips, Hide and Isolate

An edited EX-7 layout adjusted entirely with grips, plus fluent use of isolate/hide/end-isolate.

3

Lab 11 — Blocks, Attributes and Part Lists

A BALLOON block with PART_NO / DESC / QTY attributes, inserted and filled in for each part.

4

Lab 12 — External References and PDF Import

A host drawing with a live xref title block and a PDF underlay ready for tracing.

© 2026 Tertiary Infotech Academy Pte Ltd

05

TOPIC 05

Design Documentation

Text · Dimensions · Tables & Part Lists · Layouts, Viewports & Plotting

© 2026 Tertiary Infotech Academy Pte Ltd

SKILLS FRAMEWORK A5 · K5

Key Concepts — Design Documentation

1

Text & text styles

Single-line and multiline text; a text style controls font, height and justification centrally.

2

Dimensions

Linear, aligned, angular, radial, diameter, ordinate and arc-length dimensions annotate real sizes.

3

Dimension styles

A named collection of settings — arrowheads, text placement, tolerances — applied consistently.

4

Multileaders & clouds

Multileaders point notes at features; revision clouds flag changed areas for checking.

5

Tables & part lists

TABLE creates row/column data such as bills of materials extracted from block attributes.

6

Layouts & plotting

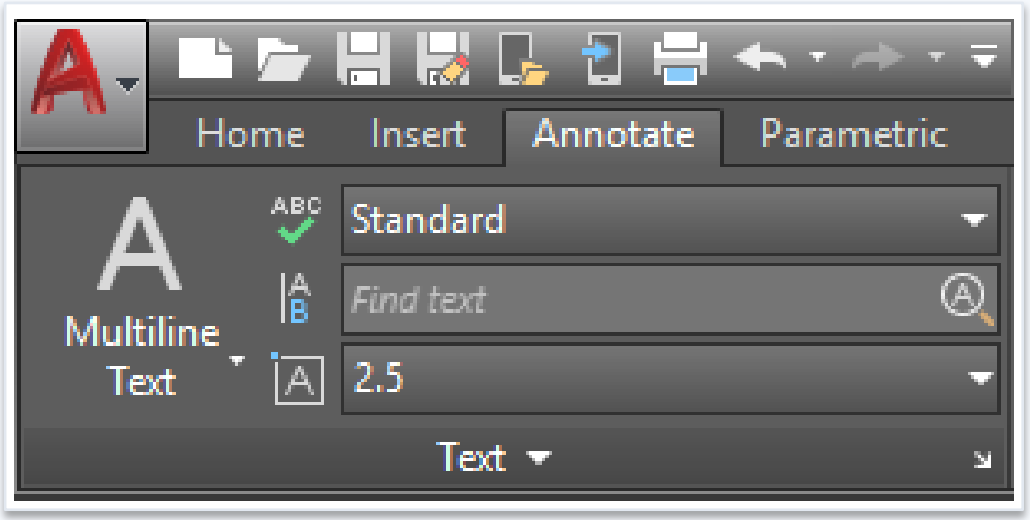
Scaled viewports on a titled layout sheet, printed with PLOT or published to PDF/DWF.

© 2026 Tertiary Infotech Academy Pte Ltd

ANNOTATION

Text and Text Styles

- Multiline text (MTEXT) fills a bounding box; single-line text (TEXT) is quick labels.
- A text style names a collection of settings: font, height, spacing, justification.
- Create styles once (STYLE) and every note updates when the style changes.
- Annotative styles size text automatically to the viewport scale.

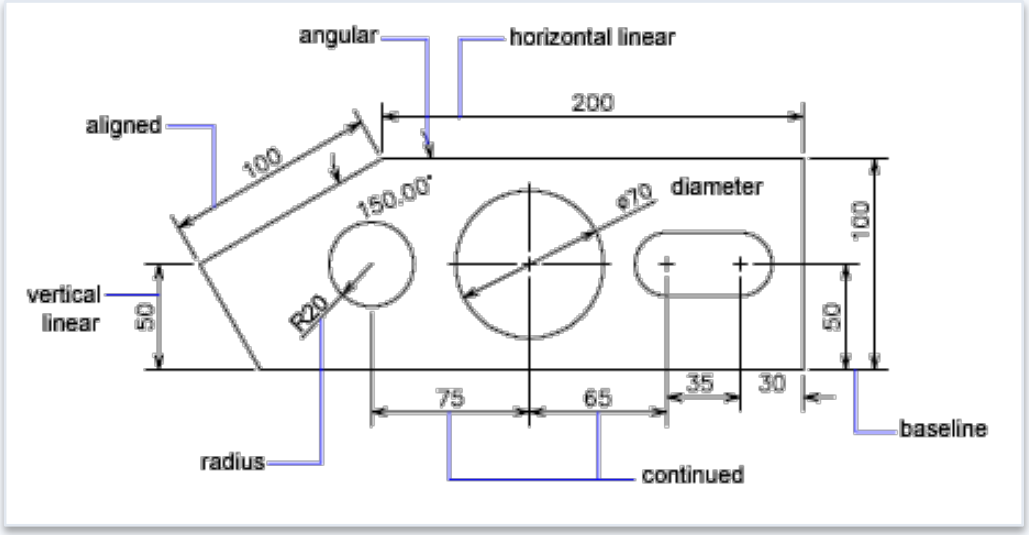


© 2026 Tertiary Infotech Academy Pte Ltd

ANNOTATION

Dimensions — The Basic Types

- Linear and aligned dimensions measure edges; angular measures between lines.
- Radial and diameter dimensions measure arcs and circles, with centre marks.
- Arc-length and ordinate dimensions cover curved lengths and datum offsets.
- Continue chains dimensions end-to-end; Baseline stacks them from one datum.



© 2026 Tertiary Infotech Academy Pte Ltd

ANNOTATION

Dimension Styles

1

What a style controls

Arrowheads, extension lines, text placement, units and tolerances — consistently.

2

Create

DIMSTYLE → New: name it, set fit, text height and precision for your standard.

3

Apply

Set the style current before dimensioning; existing dimensions can be updated.

4

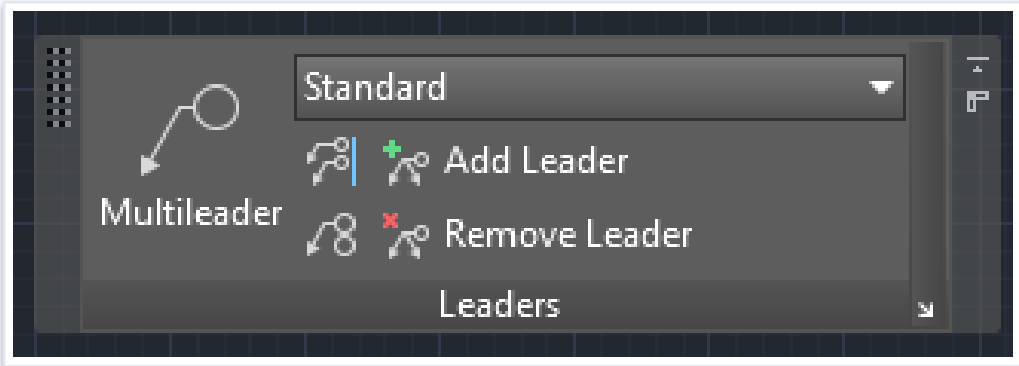
House rule

Never override individual dimension text — fix the style instead.

© 2026 Tertiary Infotech Academy Pte Ltd

Multileaders and Revision Clouds

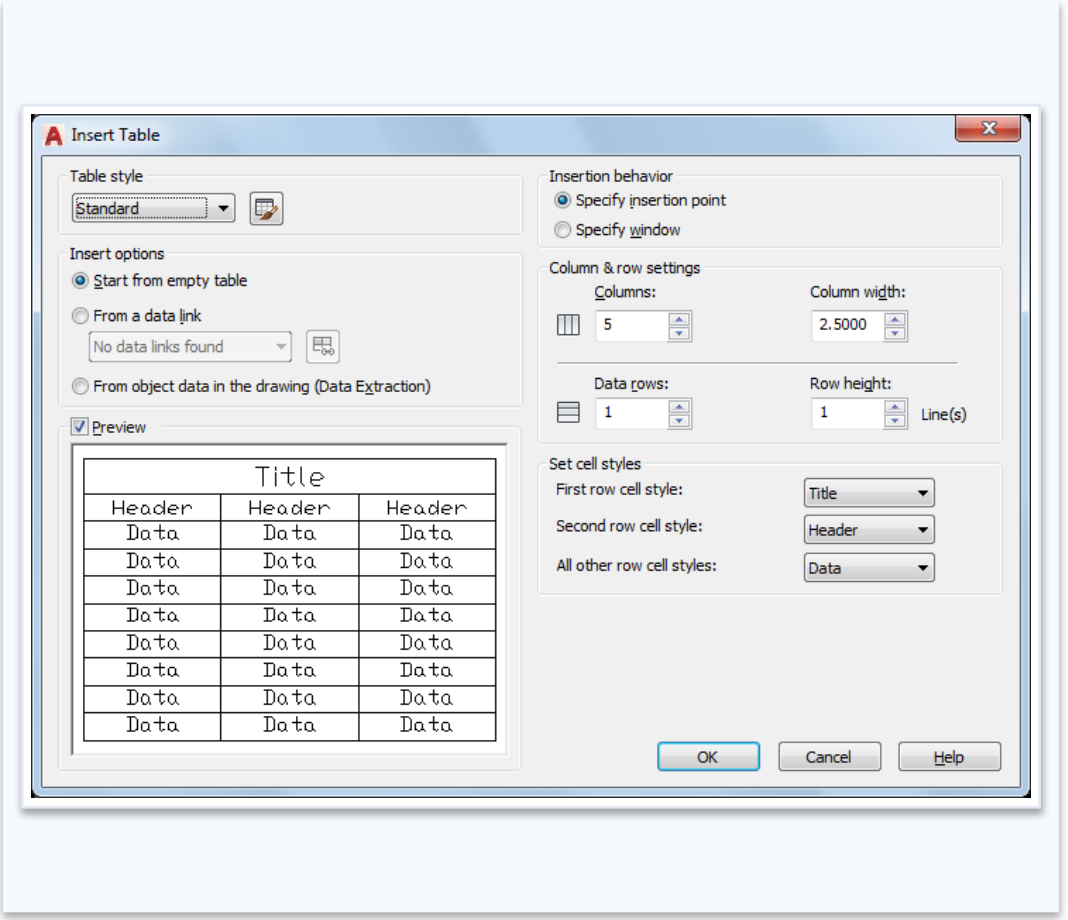
- A multileader is an arrowhead, landing line and note text (or block) in one object.
- MLEADER places it; add/remove leader lines and align multiple leaders neatly.
- Revision clouds flag changed areas for checking and approval.
- Draw a cloud by dragging, corner points — or convert an existing closed object.



© 2026 Tertiary Infotech Academy Pte Ltd

Tables and Part Lists

- TABLE creates a compound object of rows and columns from a table style.
- Type into cells like a spreadsheet; formulas and fields are supported.
- DATAEXTRACTION fills a table from block attributes — the bill of materials.
- Tables can link to external Excel data and update both ways.

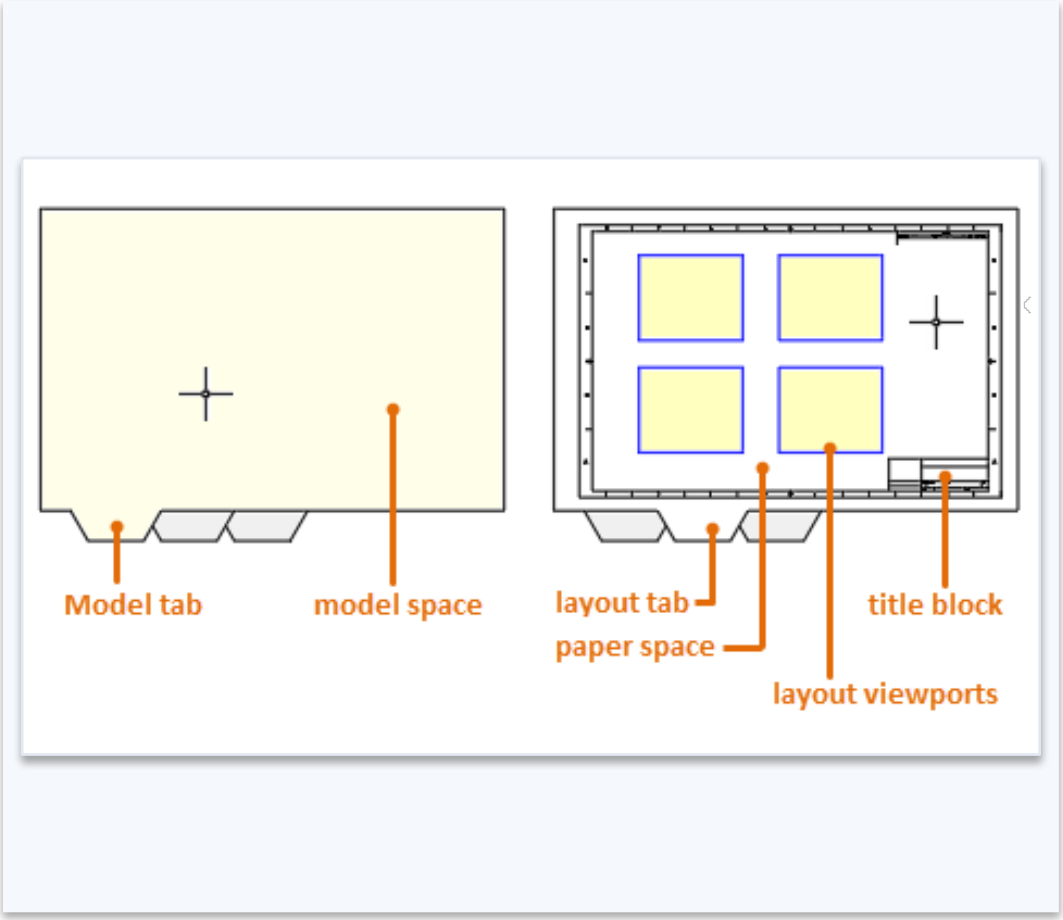


© 2026 Tertiary Infotech Academy Pte Ltd

OUTPUT

Layouts and Viewports

- A layout is the 2D sheet environment: title block, viewports, annotation.
- MVIEW creates layout viewports — windows into model space.
- Set each viewport's scale exactly (e.g. 1:20), then lock it against accidental zooming.
- One model, many sheets: each layout composes its own views and scales.

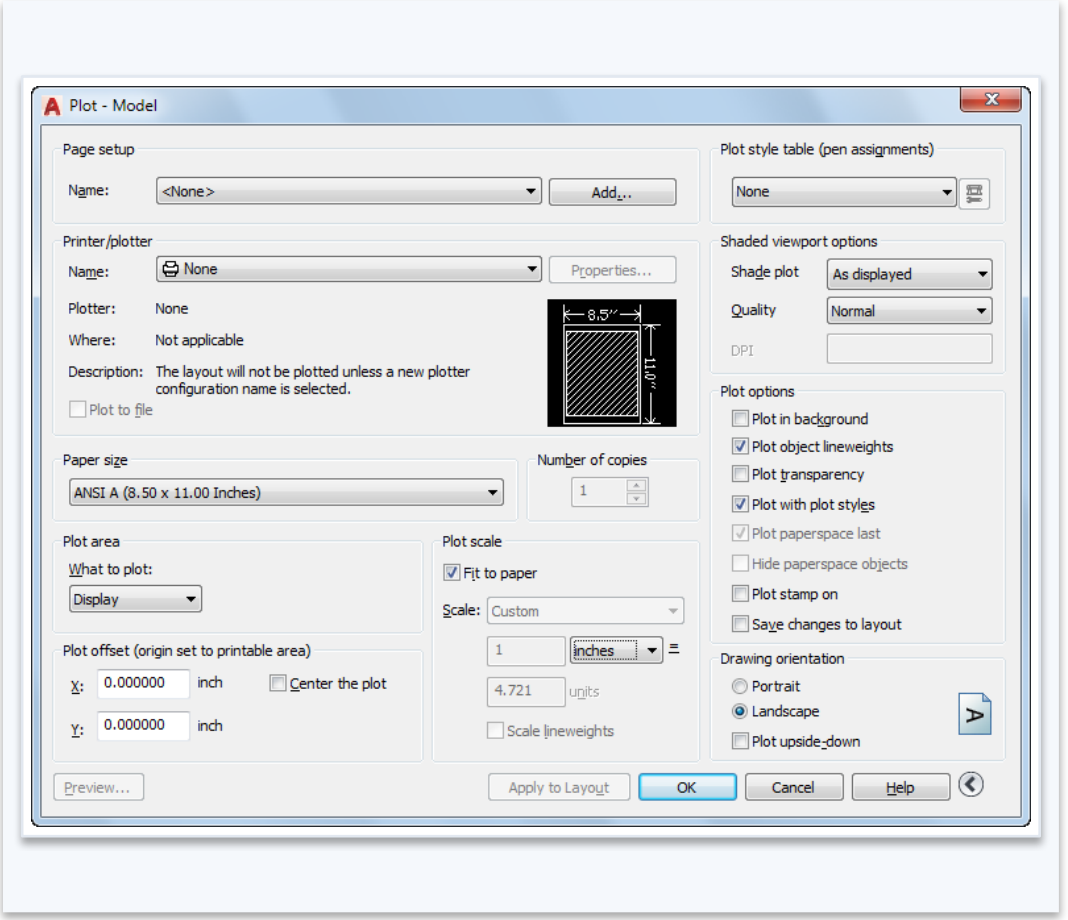


© 2026 Tertiary Infotech Academy Pte Ltd

OUTPUT

Plotting and Printing

- Plot from the layout: choose plotter, paper size, plot area, scale and orientation.
- DWG To PDF.pc3 publishes the standard submission PDF.
- Page setups save the configuration per layout and import between drawings.
- Preview before plotting — check lineweights, scale and the full sheet.

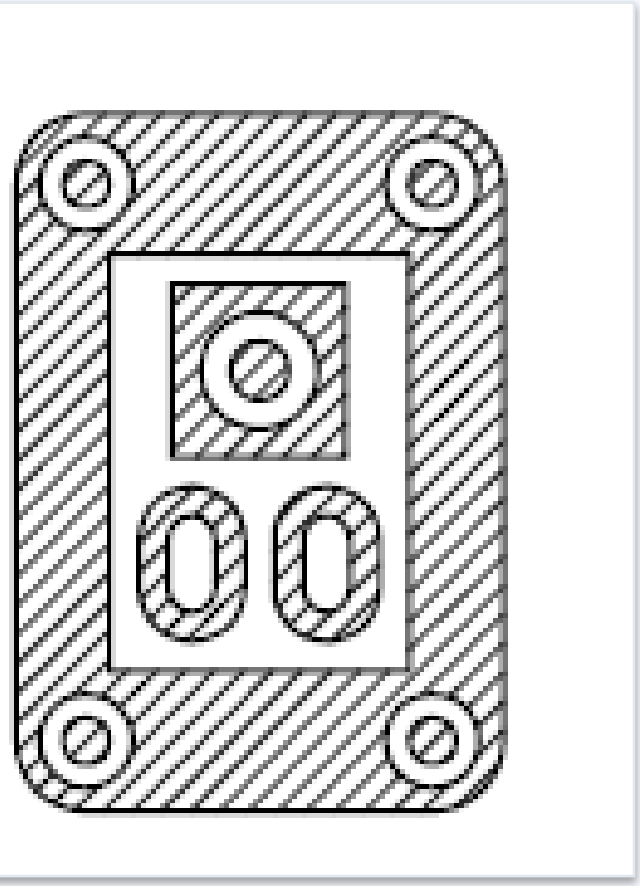


© 2026 Tertiary Infotech Academy Pte Ltd

DETAIL

Hatching and Gradients

- HATCH fills enclosed areas with patterns (ANSI31 for sections), solids or gradients.
- Pick an internal point — island detection keeps text and holes clear.
- Scale and angle make adjacent parts read distinctly in section views.
- HATCHEDIT changes an existing hatch; hatches stay associative to their boundary.



© 2026 Tertiary Infotech Academy Pte Ltd

WHAT YOU'LL DO

Hands-On Labs — Design Documentation

Labs 13–14

- Text and Text Styles
- Dimensions, Multileaders and Part-List Tables

Labs 15–16

- Hatching and Detail Finishing
- Layouts, Viewports and Plotting

—

- —

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 05 · HANDS-ON

Text and Text Styles

LAB 13

Annotate the layered table drawing with single-line and multiline text, create and apply text styles for titles and notes, and fill in the title-block fields.

You'll produce

An annotated drawing with consistent TITLE and NOTES text styles and a completed title block.

Commands: TEXT, MTEXT, STYLE, annotative text, SPELL
Files: labs/lab-13-text-and-text-styles/lab-table-lesson2.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 13 · STEP-BY-STEP

Text and Text Styles

STEPS 1–3 OF 6

1 Open your table drawing and create text styles TITLE (Arial, 5mm) and NOTES (Arial, 2.5mm)

ST → New → font/height

2 Add the drawing title with single-line text on the A-TEXT layer

DT → start point → height → rotation

3 Add a general-notes paragraph with multiline text in a bounding box

T (MTEXT) → pick corners

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 13 · STEP-BY-STEP

Text and Text Styles

STEPS 4–6 OF 6

4 Set the NOTES style current and re-style existing text via the Properties palette

```
PR → Style → NOTES
```

5 Fill in the title-block fields: title, drawn-by, date, scale, sheet number

```
double-click text → edit
```

6 Spell-check the drawing and save

```
SPELL → SAVE
```


© 2026 Tertiary Infotech Academy Pte Ltd

LAB 13 · VERIFY

Text and Text Styles

✅ Check your work

All annotation uses your named styles (none on Standard), the title block is complete, and changing the NOTES style font updates every note at once.

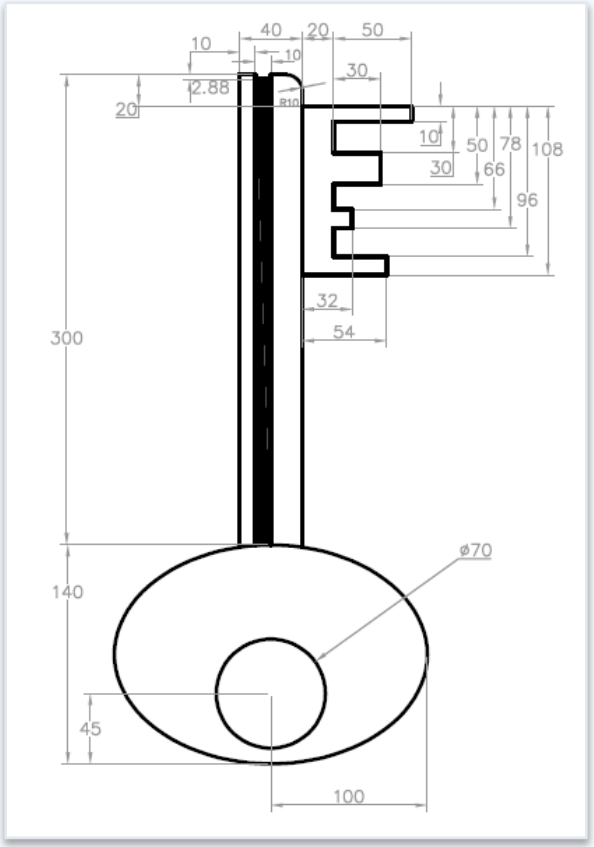
© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 05 · HANDS-ON

Dimensions, Multileaders and Part-List Tables

LAB 14

Fully dimension practice drawing EX-8 with a metric dimension style — linear, aligned, radial, diameter, angular, continue and baseline — add multileader callouts, and create the part-list table from the Lab 11 balloon attributes.



You'll produce

A fully dimensioned EX-8 sheet with leader notes and a parts table (part no · description · qty).

Commands: DIMSTYLE, DIM, DIMLINEAR/ALIGNED/RADIUS/DIAMETER/ANGULAR, DIMCONTINUE, DIMBASELINE, MLEADER, TABLE, DATAEXTRACTION

Files: labs/lab-14-dimensions-multileaders-and-part-list-tables/ex8.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 14 · STEP-BY-STEP

Dimensions, Multileaders and Part-List Tables

STEPS 1–3 OF 7

1. Open ex8.dwg and create dimension style MECH-25 (ISO, 2.5mm text, closed-filled arrows)

   D (DIMSTYLE) → New → fit/text

2. Place linear and aligned dimensions on the main edges

   DLI · DAL

3. Dimension the circles and arcs with diameter, radius and angular dimensions

   DDI · DRA · DAN

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 14 · STEP-BY-STEP

Dimensions, Multileaders and Part-List Tables

STEPS 4–6 OF 7

4 Chain related dimensions with Continue, and datum-reference them with Baseline

DCO · DBA

5 Add multileader callouts for the hole notes and edge finishes

MLD → arrow point → landing → note

6 Insert a table for the part list and populate it from the Lab 11 attribute extraction

TB (TABLE) · DATAEXTRACTION

© 2026 Tertiary Infotech Academy Pte Ltd

Dimensions, Multileaders and Part-List Tables

STEPS 7–7 OF 7

7 Add a revision cloud around one detail to flag it for checking, then save

REVCLOUD → drag around detail

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 14 · VERIFY

Dimensions, Multileaders and Part-List Tables

✅ Check your work

Every dimension on the EX-8 sheet is reproduced with the MECH-25 style (no overridden text), leaders land neatly, and the table rows match the balloons one-for-one.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 05 · HANDS-ON

Hatching and Detail Finishing

LAB 15

Apply section hatching to the closed areas of a section view — pattern, scale and angle — use gradient fills for presentation, and control hatch boundaries and islands.

You'll produce

A section view with correct ANSI/ISO hatch patterns and a presentation-filled elevation.

Commands: HATCH (pattern/scale/angle), gradient fills, boundary/island detection, HATCHEDIT
Files: labs/lab-15-hatching-and-detail-finishing/ex3.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 15 · STEP-BY-STEP

Hatching and Detail Finishing

STEPS 1–3 OF 6

1. Open a drawing with closed profiles (reuse ex3-mine.dwg) and identify the cut areas

2. Hatch the main section area with ANSI31 at a scale that reads clearly

```
H → Pattern ANSI31 → pick point
```

3. Set different angles for adjacent parts so the section reads unambiguously

```
H → Angle 90 → pick point
```


© 2026 Tertiary Infotech Academy Pte Ltd

LAB 15 · STEP-BY-STEP

Hatching and Detail Finishing

STEPS 4–6 OF 6

4 Control islands: hatch around text and holes using island detection

H → Settings → Island detection

5 Apply a two-colour gradient fill to the elevation face for presentation

GRADIENT → colours → pick area

6 Edit an existing hatch's pattern and scale, then save

HATCHEDIT → new scale

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 15 · VERIFY

Hatching and Detail Finishing

✅ Check your work

Hatches stay inside their boundaries (no bleed), adjacent parts hatch at different angles, and text/holes remain unhatched islands.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC 05 · HANDS-ON

Layouts, Viewports and Plotting

LAB 16

Compose an A3 layout: attach the title block, create scaled viewports (1:20 and a detail), lock them, and plot the sheet to PDF — the standard submission output.

You'll produce

A print-ready A3 layout with title block, two locked scaled viewports and a plotted PDF.

Commands: LAYOUT, PAGESETUP, MVIEW, viewport scale/lock, PLOT, DWG to PDF.pc3
Files: labs/lab-16-layouts-viewports-and-plotting/title-block.dwg

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 16 · STEP-BY-STEP

Layouts, Viewports and Plotting

STEPS 1–3 OF 6

1. Open your annotated drawing, switch to a Layout tab and rename it A3-SHEET

   right-click tab → Rename

2. Set the page: DWG To PDF.pc3, ISO A3 (420×297), landscape, 1:1 plot scale

   PAGESETUP → Modify

3. Insert the title block on the layout at 0,0 in paper space

   I → title-block

© 2026 Tertiary Infotech Academy Pte Ltd

LAB 16 · STEP-BY-STEP

Layouts, Viewports and Plotting

STEPS 4–6 OF 6

4 Create a main viewport and set its scale to 1:20 exactly, then lock it

```
MV → drag → select → scale 1:20 → lock
```

5 Add a second, smaller viewport for a detail at a larger scale

```
MV → drag → scale 1:5
```

6 Plot the layout to PDF and check line weights in the output

```
PLOT → DWG To PDF.pc3 → OK
```


© 2026 Tertiary Infotech Academy Pte Ltd

LAB 16 · VERIFY

Layouts, Viewports and Plotting

✅ Check your work

Both viewports report their exact scales in the status bar and cannot be zoomed when locked, and the plotted PDF shows the complete titled sheet at A3.

© 2026 Tertiary Infotech Academy Pte Ltd

TOPIC RECAP

Recap — Design Documentation

1

Lab 13 — Text and Text Styles

An annotated drawing with consistent TITLE and NOTES text styles and a completed title block.

2

Lab 14 — Dimensions, Multileaders and Part-List Tables

A fully dimensioned EX-8 sheet with leader notes and a parts table (part no · description · qty).

3

Lab 15 — Hatching and Detail Finishing

A section view with correct ANSI/ISO hatch patterns and a presentation-filled elevation.

4

Lab 16 — Layouts, Viewports and Plotting

A print-ready A3 layout with title block, two locked scaled viewports and a plotted PDF.

© 2026 Tertiary Infotech Academy Pte Ltd

WRAP-UP

Course Summary & Next Steps

© 2026 Tertiary Infotech Academy Pte Ltd

LEARNING OUTCOMES

What You Achieved

1

Getting started

Navigated the AutoCAD UI, workspaces, units and precision drafting aids.

2

Technical drawing

Produced accurate 2D drawings and isometric views from dimensioned sheets.

3

Modifying objects

Built designs efficiently with move/copy, offset, trim, fillet and arrays.

4

Organizing objects

Structured drawings with layers, blocks, attributes and external references.

5

Design documentation

Dimensioned, annotated, tabulated part lists and plotted submission-ready sheets.

© 2026 Tertiary Infotech Academy Pte Ltd

TEST YOURSELF · EXAMS.TERTIARYINFOTECH.COM

Practice Exam

- Sharpen your readiness with the Tertiary Infotech AutoCAD practice exam.
- Go to exams.tertiaryinfotech.com → Practice Exams → Autodesk → AutoCAD.
- Attempt it under timed conditions and review every explanation.
- Revisit any lab whose topic you miss, then re-take the practice exam.



© 2026 Tertiary Infotech Academy Pte Ltd

KEEP LEARNING

Recommended Next Courses

1. **Architecture Drawing with Revit**
   WSQ · Tertiary Infotech Academy

2. **Product Design with Fusion 360**
   WSQ · Tertiary Infotech Academy

3. **AutoCAD Civil 3D for Infrastructure Design**
   WSQ · Tertiary Infotech Academy

4. **Technical Drawing with AutoCAD Mechanical**
   WSQ · Tertiary Infotech Academy

5. **Technical Drawing with AutoCAD Electrical**
   WSQ · Tertiary Infotech Academy

© 2026 Tertiary Infotech Academy Pte Ltd

WE'RE HERE TO HELP

Support

- If you have any enquiries during and after the class, contact us:
- Email: enquiry@tertiaryinfotech.com
- Tel: +65 6100 0613
- Website: www.tertiarycourses.com.sg

© 2026 Tertiary Infotech Academy Pte Ltd

BEFORE YOU LEAVE

Cert & TRAQOM Survey (Mandatory)

- Complete the TRAQOM course-quality survey on the LMS — it is mandatory for WSQ funding.
- Your e-certificate is issued on the LMS after you are assessed Competent and the survey is done.
- Portal: https://lms-tms.tertiaryinfotech.com

© 2026 Tertiary Infotech Academy Pte Ltd

ASSESSMENT

Final Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 1 hour, open book.
- Practical Performance (PP) — produce a dimensioned AutoCAD technical drawing, 1.5 hours, open book.
- Open book: slides, Learner Guide and approved materials only.
- Remember to take the Assessment digital attendance (TRAQOM).
- Submit your completed answers and DWG files on the LMS at https://lms-tms.tertiaryinfotech.com.

© 2026 Tertiary Infotech Academy Pte Ltd

ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit WA (SAQ) then Practical Performance — open book
4. Submit your answers and DWG file on the LMS
5. Sign the Assessment Summary Record

© 2026 Tertiary Infotech Academy Pte Ltd

Digital Attendance (Mandatory)

- It is mandatory to take the AM, PM and Assessment digital attendance for WSQ-funded courses.
- The trainer/administrator displays the digital attendance QR code from the SSG portal.
- Scan the QR code with your mobile phone camera and submit your attendance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

© 2026 Tertiary Infotech Academy Pte Ltd

HAPPY DRAFTING

Thank You!

You are now ready to read, produce and document technical drawings with AutoCAD.

© 2026 Tertiary Infotech Academy Pte Ltd